Lecture 14:

Language Models and Transformer

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Machine Learning Algorithms

MSSE 277B, 3 Units

Course Map

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classic ML tools & algorithms

Lecture 2: Bayesian Methods in Machine Learning

Lecture 3: Dimensionality Reduction: Principal Component Analysis

Lecture 4: Linear and Non-linear Regression and Classification

Lecture 5: Unsupervised Learning: K-Means, GMM, Trees

Lecture 6: Adaptive Learning and Gradient Descent Optimization Algorithms

ANNs/AI/Deep Learning

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Lecture 9: Convolutional Neural Networks (CNNs) - Part I

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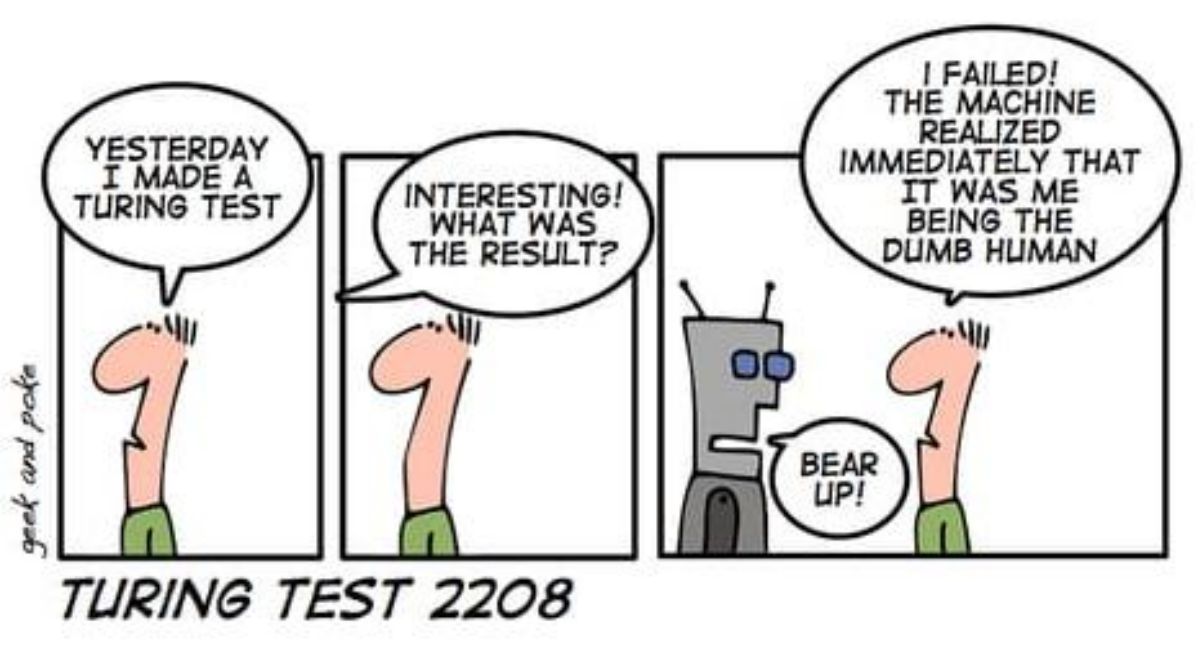
Lecture 12: Combining LSTMs and CNNs

Lecture 13: Project Presentations

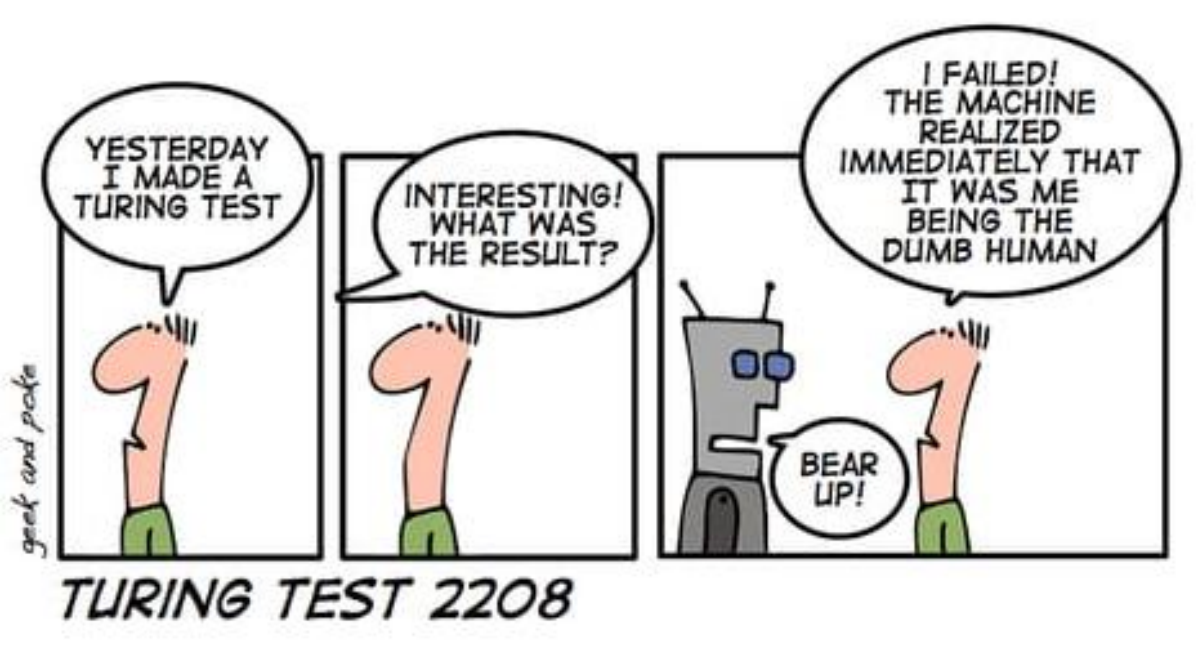
Lecture 14: Transformer

Lecture 15: GNN



Outline

- **Introduction**
- **Bigram and MAP**
- **Positional Encoding**
- **Word Embedding**
- **Attention**
- **Transformer Architecture**



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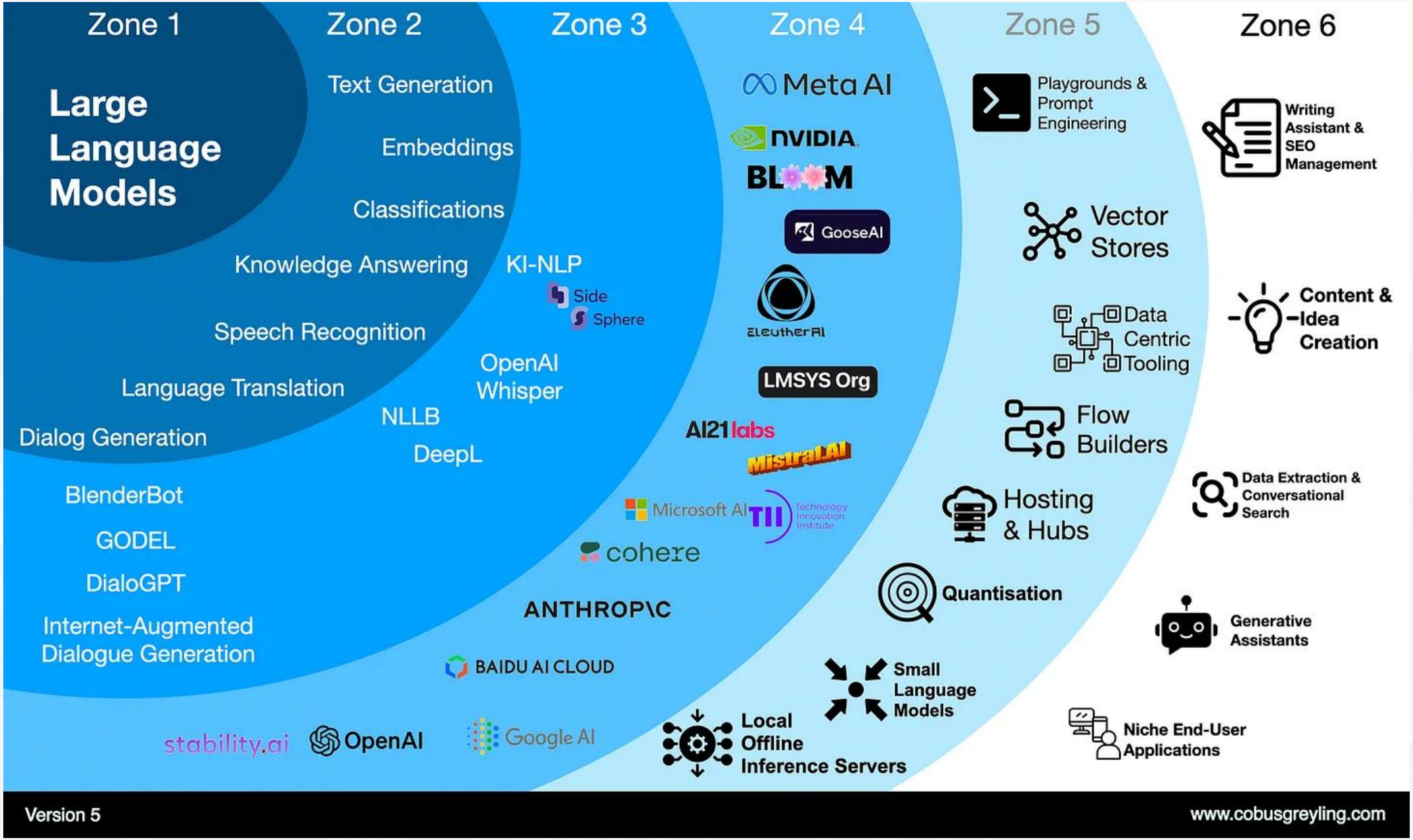
LLM & Transformer:



LLM & Transformer:

corpus: (large, representative) data set containing sequences of a language

token: individual, independent entity of a language

alphabet/vocabulary: set of tokens

token	**size of *alphabet***
- letters in a word	- 10^2
- words in a sentence	- $10^4 \ldots 10^6$
(**upper/lower case, cases, gender, tenses, conjugations**)	
- amino acids in a protein sequence	- 21
- nucleotides in a DNA/RNA sequence	- 4
- motifs in a DNA/RNA sequence	- 10^4

LLM & Transformer:

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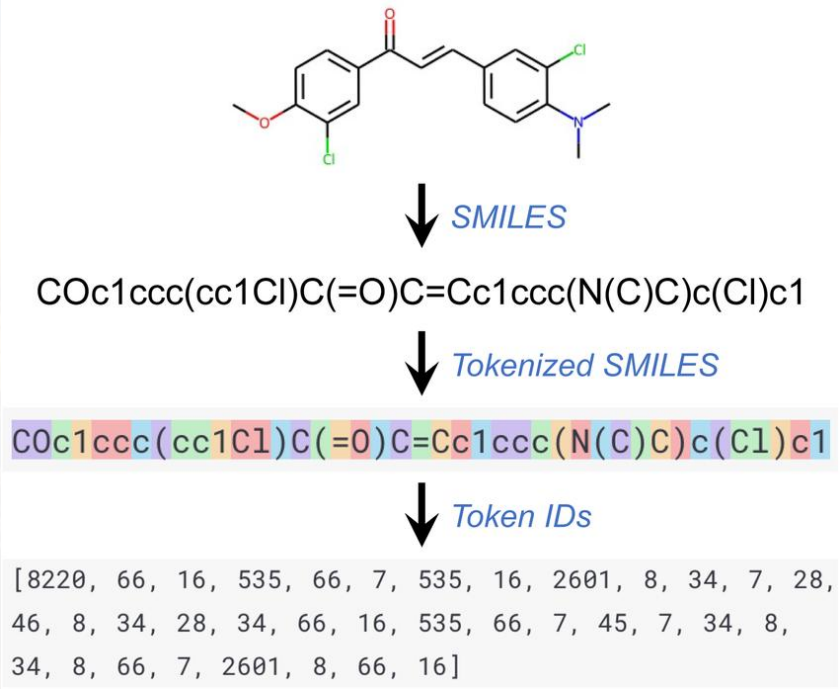
tokenization



token:
- single atom vs…
- …functional group

DOI:10.1039/D3SC04610A

LLM & Transformer:

corpus: (large, representative) data set containing sequences of a language

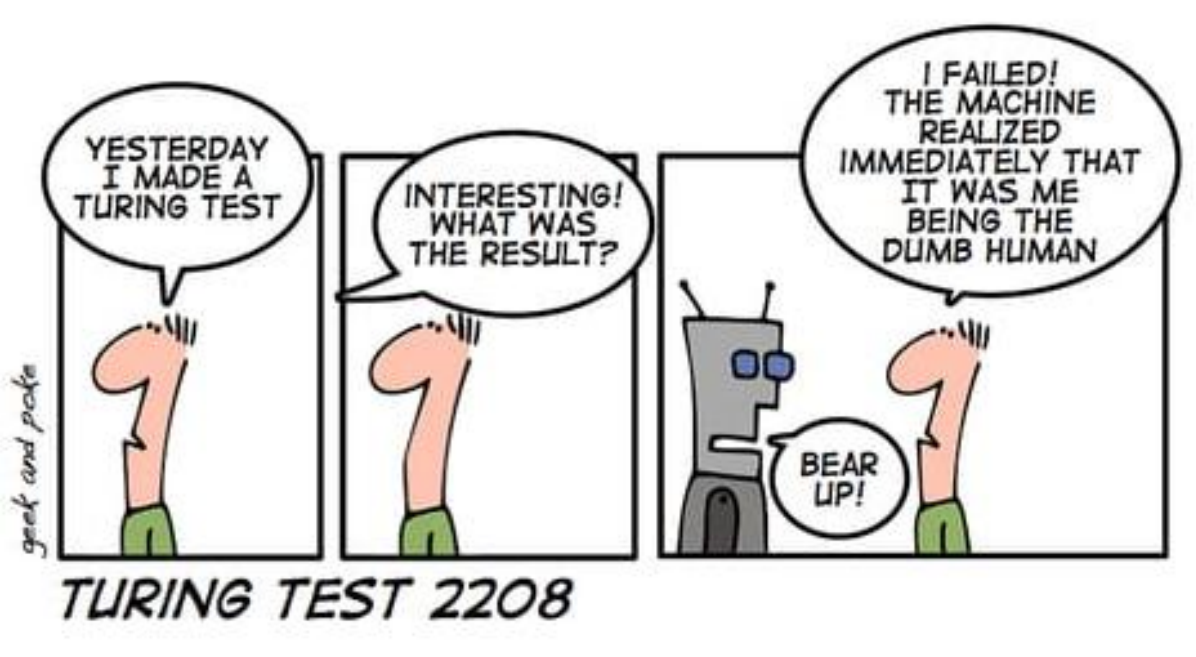
token: individual, independent entity of a language

alphabet/vocabulary: set of tokens

note: language models don't know grammar as we do, but they don't need to anyway…

three things make context (details: see later**):**

- **word embedding** (relation between similar/different token)
- **positional encoding** (location of token in a sequence)
- **attention** (relation between token within a sequence)



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LLM & Transformer:

$X_1\ X_2\ X_3\ X_4\ X_5 \ldots . X_n$ sequence of *n* **token** *X*

<u>actually:</u>

$$P(X_1\ X_2\ X_3\ X_4\ X_5 \ldots . X_n) = P(X_n|X_{n-1} \ldots X_1)P(X_{n-1}|X_{n-2} \ldots X_1) \ldots P(X_1)$$

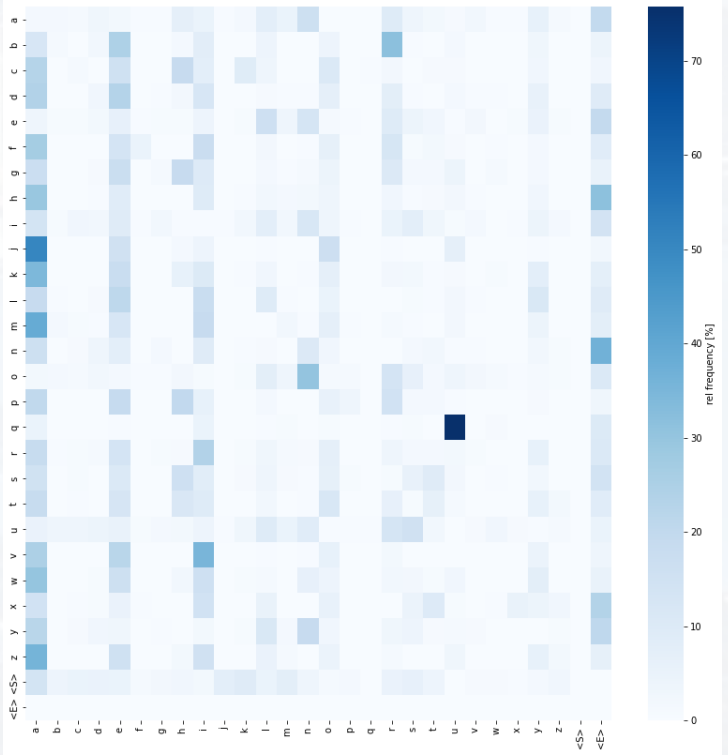
<u>bigram</u> <u>(1st order Markov Chain, see e.g. first WhatsApp versions):</u>

$$P(X_1\ X_2\ X_3\ X_4\ X_5 \ldots . X_n) = P(X_n|X_{n-1})P(X_{n-1}|X_{n-2}) \ldots P(X_1)$$

P(i|j): that token ***i*** is generated after token ***j***

→ N x N transition matrix from frequencies

→ **"bigram" = "two words"**



frequency matrix of letters in common names

LLM & Transformer:

bigram (1st order Markov Chain):

let's build our own bigram model: **generate new names** based on a corpus of names

```
In [15]: words[0:12]
Out[15]:
['emma',
 'olivia',
 'ava',
 'isabella',
 'sophia',
 'charlotte',
 'mia',
 'amelia',
 'harper',
 'evelyn',
 'abigail',
 'emily']
```

see **Andrej Karpathy's GitHub** repository

$$P(X_1\, X_2\, X_3\, X_4\, X_5 \ldots . X_n) = \boxed{P(X_n|X_{n-1})P(X_{n-1}|X_{n-2})} \ldots \boxed{P(X_1)}$$

we only need to count **how often** a letter is followed by another

we also need to indicate when a name has **started** and **ended**

```
['<S>'] + ['olivia'] + ['<E>']
```

→ **alphabet:** 26 letters + the two special "letters"

let's create a dictionary first (will help for counting):

LLM & Transformer:

bigram (1st order Markov Chain):

let's build our own bigram model: **generate new names** based on a corpus of names

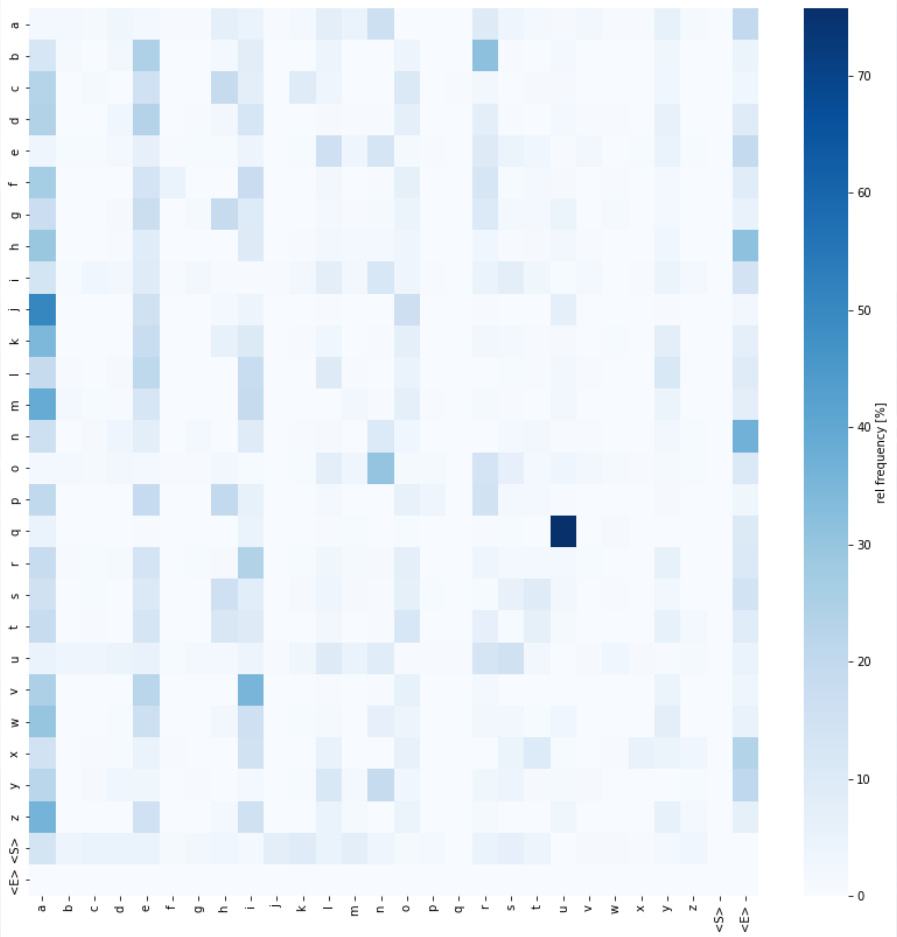
1) dictionary first (will help for counting)
2) count how often letter ***i*** is followed by letter ***j***

   → bigram matrix ***N***

3) normalize ***N*** accordingly
4) begin with a start token
5) draw a letter randomly based on ***N***, using

```
np.argmax(np.random.multinomial(1,p))
```

6) if next token is stop token → stop



LLM & Transformer:

bigram (1st order Markov Chain):

let's build our own bigram model: **generate new names** based on a corpus of names

check out **`Bigram.ipynb`**

`B.SampleNames(15)` vs totally random **`B.SampleNames(15, False)`**

some names are gibberish

some names sound real

some names **are** real

```
In [295]: B.SampleNames(15)
keesa
ann
ja
jon
nma
malynojana
sall
daha
drvah
lzaxi
tyunusthun
jorrwro
ja
asoow
s
```

```
In [296]: B.SampleNames(15,False)

mtkgy
yufexhviovmorhqvikbbbjxebpxwurejaqlzzuwuanxmmomhr<S>uhb
xlmusadjfdzxadaotd
ik<S>vdtydvxevtaselkykcfbamceprtvl
zyr<S>inzoerobzwuovx
eg<S>pbdvikf<S>tomcnkfsjay<S>rikatnaykizszcivpds<S>zj
kh<S>y<S>ualzugqgakakeubjbasc
bblupnibtqmyl<S>vyobf
kybs
rznjgpmlo
tnhoxuckkjjzbwmj<S>vshkycicf<S>kowskphy
rxodh
jvswmzw
jzpcfnpcg
```

Note, there is no conceptual difference between applying our model to *letters in a word* vs *words in a sentence*

<u>caveats:</u>

- the bigram model derives $P(X_n)$ from **observed** frequencies
  → essentially **MLE** (problematic if a letter hasn't appeared in the sequence yet
  → $P(X_n)$ assumed to be zero!)

```
Nsam = N/np.sum(N+0.0001, axis = 1, keepdims = True)
S_bi += np.sum(-N[:,i]*np.log(N[:,i]+1e-16))
```

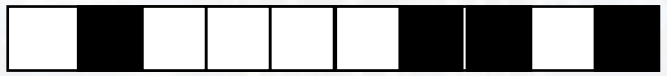
- can we implement something that is closer to:

$$P(X_1\, X_2\, X_3\, X_4\, X_5 \ldots . X_n) = P(X_n | X_{n-1} \ldots X_1) P(X_{n-1} | X_{n-2} \ldots X_1) \ldots P(X_1) \quad ?$$

binomial process

$$P(k|n,q) = \binom{n}{k} q^k (1-q)^{n-k}$$

$1-q$ q



$q = ?$

LLM & Transformer:

$$P(X_1\, X_2\, X_3\, X_4\, X_5 \ldots . X_n) = P(X_n | X_{n-1} \ldots X_1) P(X_{n-1} | X_{n-2} \ldots X_1) \ldots P(X_1)$$

Bayesian
Parameter
Estimation*)

$1 - q$ $\qquad$ q

$$P(k|n,q) = \binom{n}{k} q^k (1-q)^{n-k} \qquad q = ?$$

likelihood function **(here: binomial)**

$$P(q|data\ set) = \frac{\boxed{P(data\ set|q)}\boxed{P(q)}}{\boxed{P(data\ set)}}$$

prior (~$P(X_n|X_{n-1} \ldots X_1)$)

evidence **(const wrt q)**

$q = const$
before 1st data point
(max entropy!)

$$= \frac{1}{\int_0^1 P(q|data\ set) dq} (1-q)^{n-k} q^k$$

q = conjugate prior
after nth data point

$$= \frac{q^{k+\alpha-1}(1-q)^{n-k+\beta-1}}{\int_0^1 q^{k+\alpha-1}(1-q)^{n-k+\beta-1}\, dq}$$

***) see lecture 2**

LLM & Transformer:

Bigram & MAP

$$P(X_1\, X_2\, X_3\, X_4\, X_5 \ldots . X_n) = P(X_n | X_{n-1} \ldots X_1) P(X_{n-1} | X_{n-2} \ldots X_1) \ldots P(X_1)$$

$$P(k|n,q) = \binom{n}{k} q^k (1-q)^{n-k} \qquad q = ?$$

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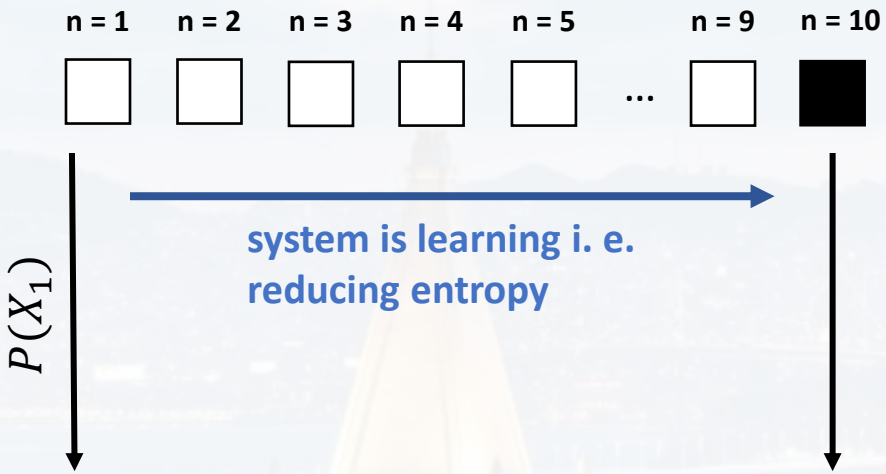
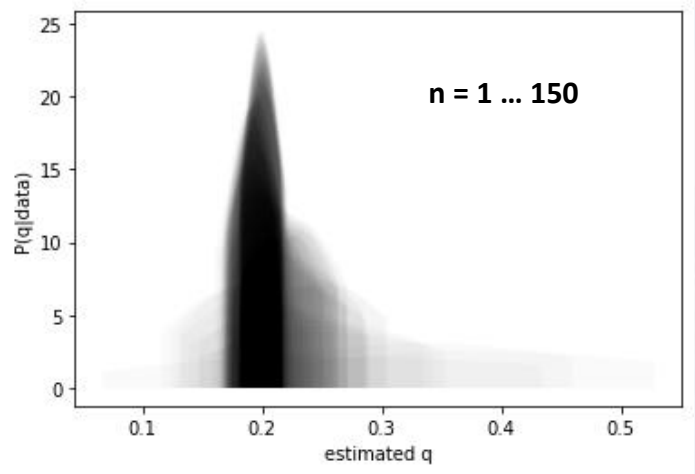
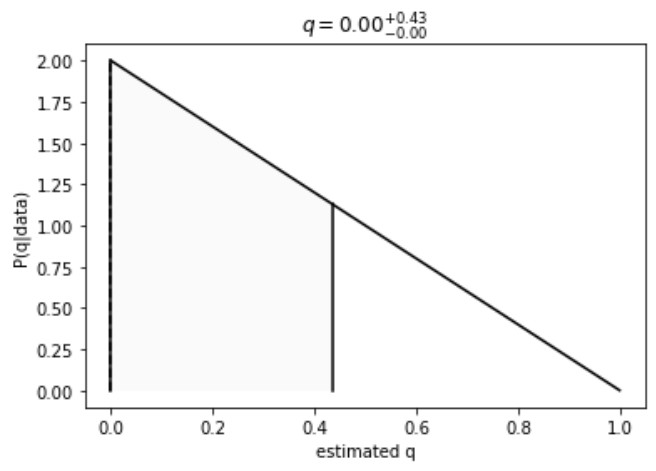
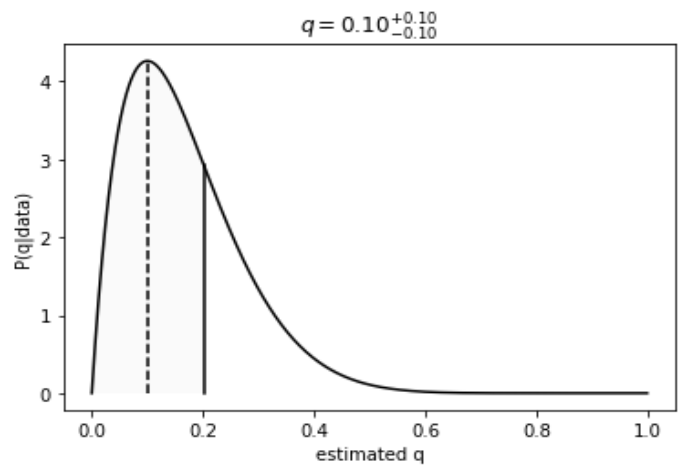


$P(X_1\, X_2\, X_3\, X_4\, X_5 \ldots . X_n)$







*) see lecture 2

LLM & Transformer:

Bigram & MAP

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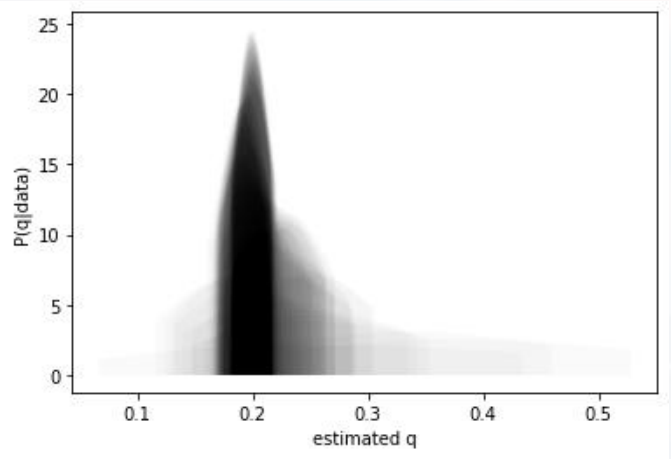
$$P(k|n,q) = \binom{n}{k} q^k (1-q)^{n-k}$$

$$P(q|data\ set) = \frac{q^{k+\alpha-1}(1-q)^{n-k+\beta-1}}{\boxed{\int_0^1 q^{k+\alpha-1}(1-q)^{n-k+\beta-1}\, dq}}$$

Beta function



more general, we want to learn the probability $P_j(a)$ of letter a at position j $\quad q \rightarrow P_j(a)$

→ multinomial problem

→ conjugate prior is the **Dirichlet distribution**

$$P(sequence) \sim \prod_j \boxed{\prod_a P_j(a)^{\alpha(a)-1}}$$

equivalent to what was $P(q|data\ set)$ earlier

***) see lecture 2**

LLM & Transformer:

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Dirichlet distribution

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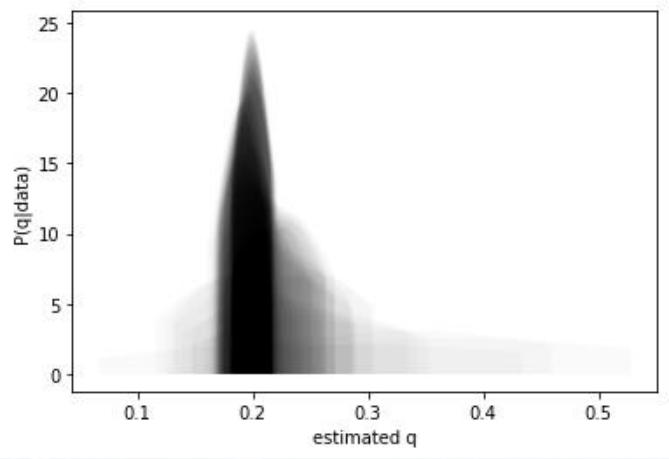

note: $\sum_{over\ all\ a} P(a)_j = 1$ → N – dim simplex

note:

- we don't need to extract $P(a)$ from the maximum of the pdf given by the BPE posterior

- we can directly derive the maximum of $P(a)$ from $P(sequence)$ given the constrain $\sum_{over\ all\ a} P(a)_j = 1$ **(Lagrangian multipliers)**

- **Maximum a-posteriori (MAP)** approach → see XXmotif (Siebert & Soeding, 2016)

LLM & Transformer:

$$P(X_1\, X_2\, X_3\, X_4\, X_5 \ldots . X_n) = P(X_n | X_{n-1} \ldots X_1) P(X_{n-1} | X_{n-2} \ldots X_1) \ldots P(X_1)$$

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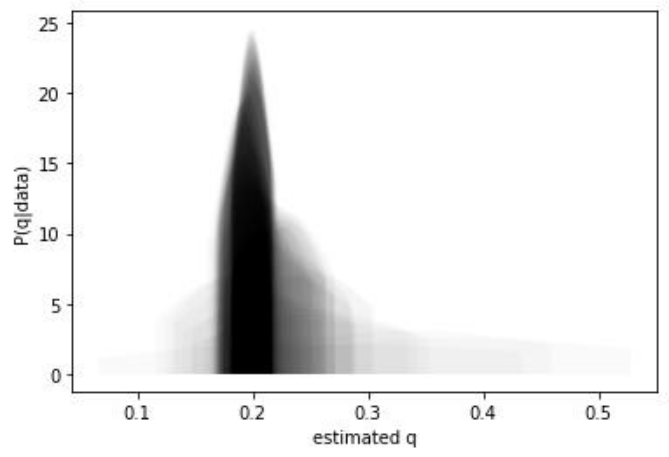

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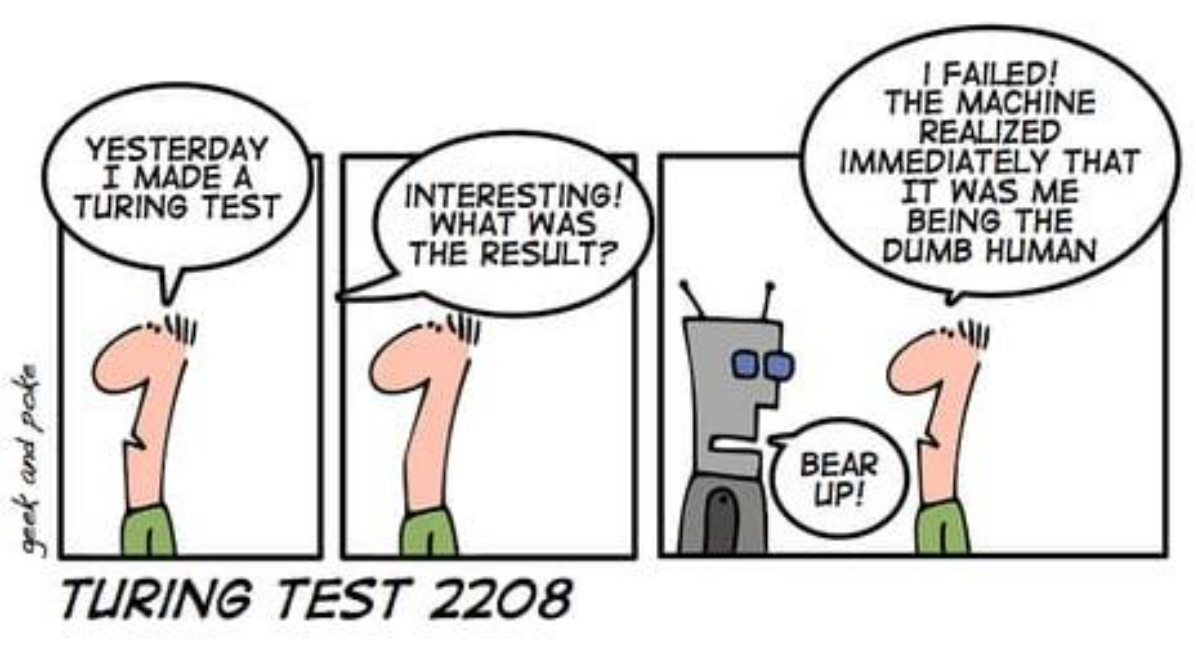
Maximum a-posteriori (MAP)

- XXmotif (Siebert & Soeding, 2016) significantly outperformed PWMs

- it struggled however with related motifs which where **physically located far apart** from each other

→ solution see later: attention
→ older solutions: LSTMs



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LLM & Transformer:

three things make context:

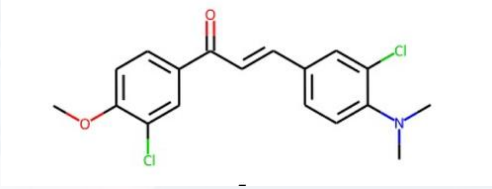
- **positional encoding** (location of token in a sequence)
- **word embedding** (relation between similar/different token)
- **attention** (relation between token within a sequence)

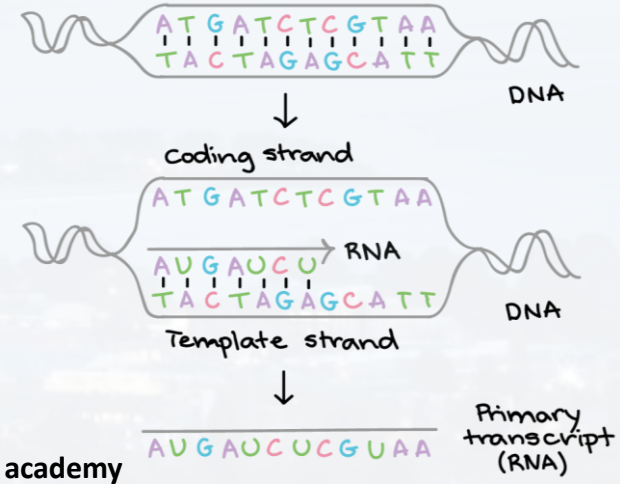
"The cat jumped on the roof."

order matters!:

1st: article
2nd: noun/subject
3rd: verb
4th : noun/object (in English)

→ **positional encoding**





Khan academy

LLM & Transformer: Positional Encoding

goal: find a positional encoding that is

- reasonably simple
- independent from the length of the sequence
- somehow normalized

<u>one idea: n-bit binary encoding</u>

position	code
	76543210 ← 8bit i.e. eight dimensions
1	00000001
2	00000010
3	00000011
4	00000100
5	00000101
6	00000110
7	00000111
8	00001000
9	00001001
10	00001010
11	00001011
12	00001100
13	00001101
14	00001110
15	00001111
16	00010000

depending on dimensions (bit)

→ different frequencies

bit	**frequency**
0	1/2
1	1/4
2	1/8
…	

Does that look familiar?
→ *like* Fourier Series

even dimensions: $E(p, 2k) = \sin\left(\frac{p}{10{,}000^{\frac{2k}{d}}}\right)$

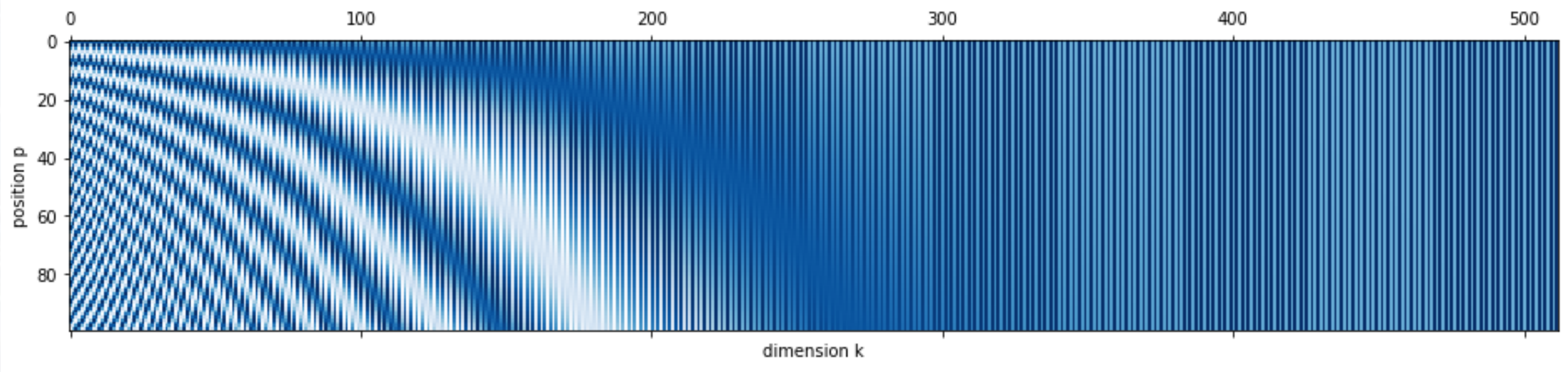
odd dimensions: $E(p, 2k+1) = \cos\left(\frac{p}{10{,}000^{2k/d}}\right)$

p: position in sequence
k: dimension index
d: number of dimensions
10,000: an arbitrary number

Vaswani et al., 2017

run **`PlotPositionEncoding.py`**

more info here



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p: position in sequence
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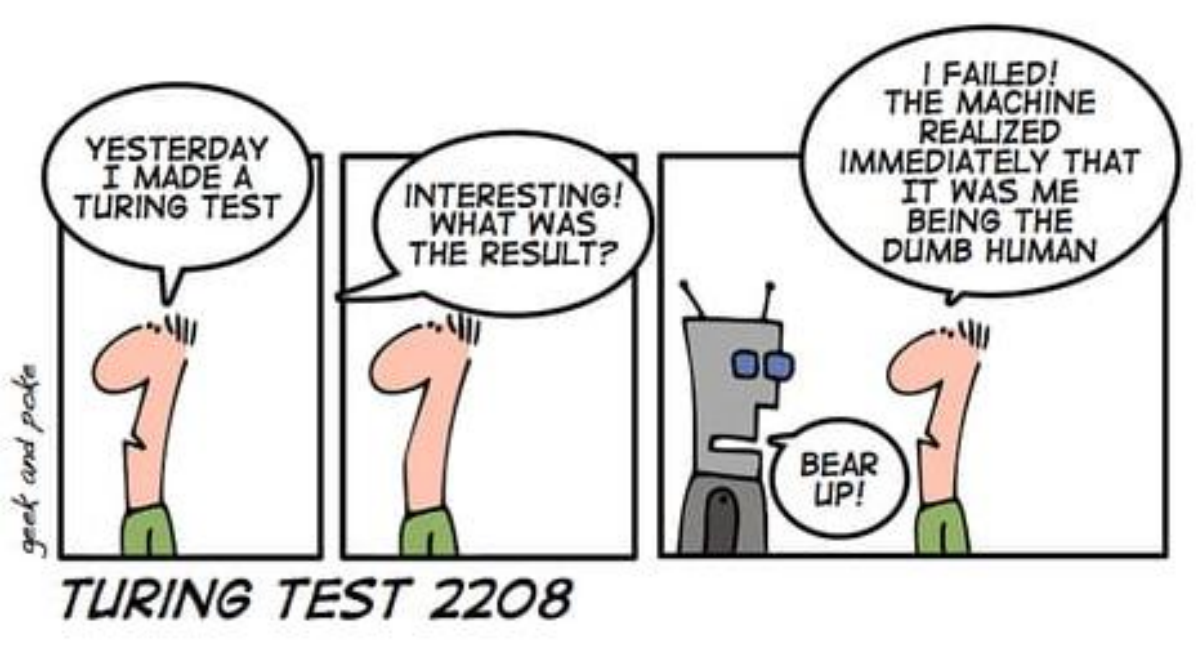
Vaswani et al., 2017

run **`PlotPositionEncoding.py`**

note:

- easier to handle numerically vs discrete encoding

- $\cos(x + 2\pi k) = \cos(x)$ and $\sin(x + 2\pi k) = \sin(x)$
  → absolute position not relevant but ***relative*** position



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problem: turning token (words/letters) into numbers

<u>single letters:</u>

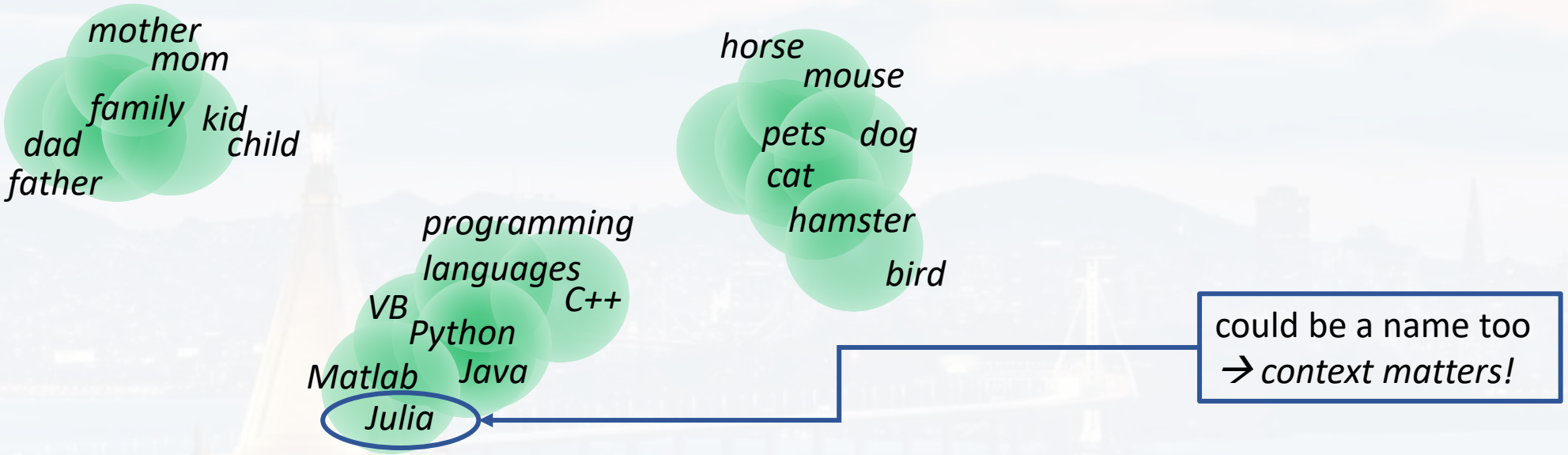
ACGT - one – hot works perfectly (four different token)
abcd… - lower/upper case, special characters (50 different token), one – hot is fine too

<u>words:</u>

actual words - 10^4 … 10^6 (upper/lower case, cases, gender, tenses, conjugations)
→ one – hot doesn't work (matrices would be too large)
→ some words have a **similar meaning,** should be **close** in parameter space (**cluster**)

LLM & Transformer: Word Embedding

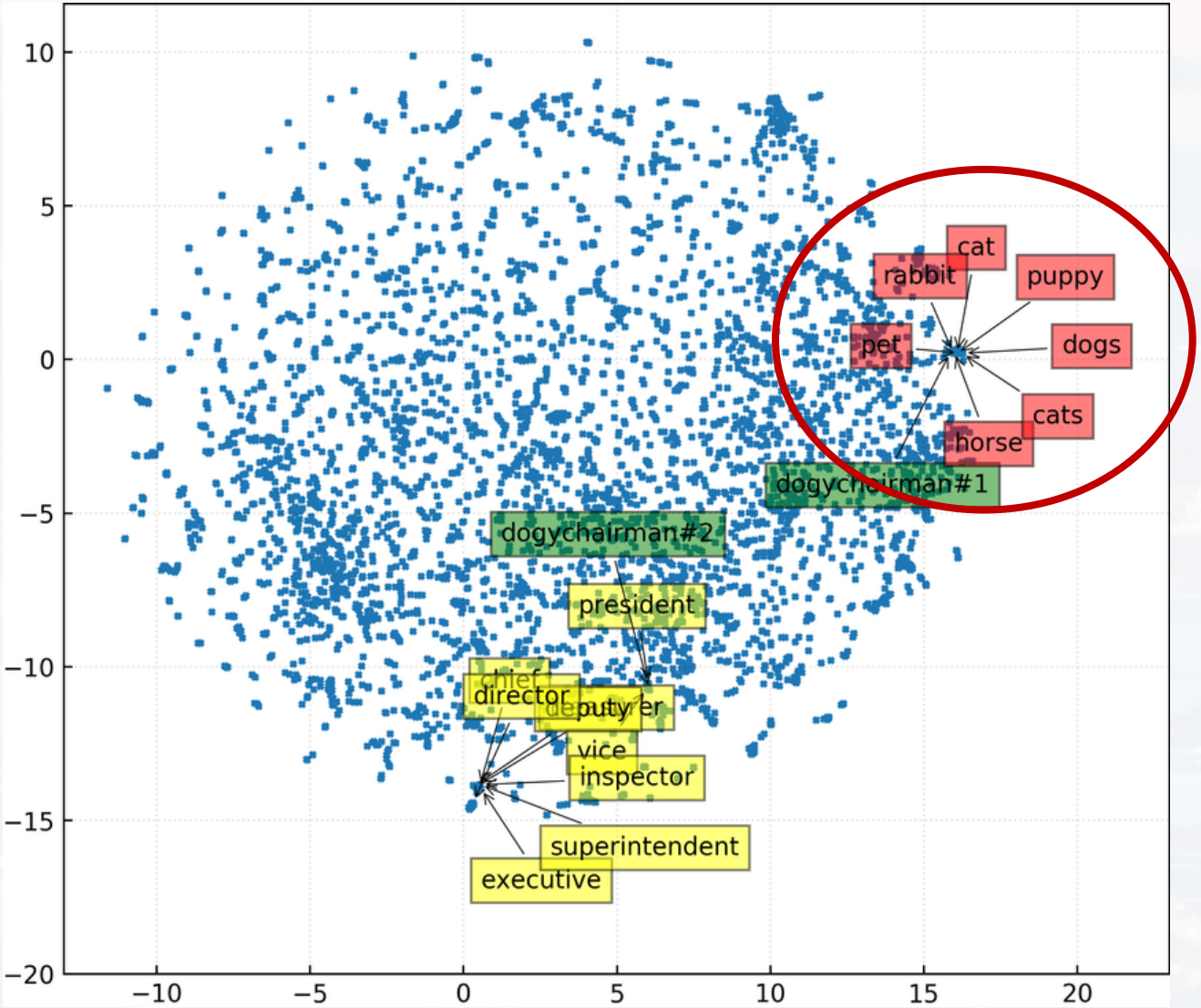
words with a **similar meaning** should form **cluster**



- embedding, instead of one – hot encoding
- from experience**: N = 30 – 300** dim vector for each token **(which is a lot less than 10^4 … 10^6)** is sufficient
- as a result: token with **similar meaning are close** in the vector space!

LLM & Transformer:

words with a **similar meaning** should form **cluster**



"Joint Learning of Sense and Word Embeddings"
M Alsuhaibani & D Bollegala

LLM & Transformer:

common training set: recorded speeches from the European Parliament:

…It seems absolutely disgraceful that we pass legislation and do not adhere to it ourselves. Mrs Lynne, you are quite right and I shall check whether this has actually not been done. I shall also refer the matter to the College of Quaestors, and I am certain that they will be keen to ensure that we comply with the regulations we ourselves vote on.
Madam President, Mrs Díez González and I had tabled questions on certain opinions of the Vice-President, Mrs de Palacio, which appeared in a Spanish newspaper.
The competent services have not included them in the agenda on the grounds that they had been answered in a previous part-session.
I would ask that they reconsider, since this is not the case….

words of similar meaning should appear in similar environment

→ *target token within a window*

Two common algorithms are ***C**ontinuous **B**ag **O**f **W**ords* and ***skip gram***

LLM & Transformer:

***C**ontinuous **B**ag **O**f **W**ords*

n:	**number of unique token from corpus**
m:	**desired number of dimensions for embedding**

The competent services have not included them in the agenda on the grounds that they…

target: *have*

context window: *The competent services not included them*

one – hot encoding

one – hot encoding

shallow ANN:

- input layer of shape *n x m*
- hidden layer of shape *m x n*
- softmax

output vector of length *n*

target vector of length *n*

learning weights such that both vectors **are identical**

→ **weights*[:,i]*** of the **hidden layer** make up the embedding vector of ***token i***

LLM & Transformer:

***C**ontinuous **B**ag **O**f **W**ords*

n:	**number of unique token from corpus**
m:	**desired number of dimensions for embedding**

The competent services have not included them in the agenda on the grounds that they…

target: *not*

context window: *competent services have included them in*

one – hot encoding

one – hot encoding

shallow ANN:

- input layer of shape $n \times m$
- hidden layer of shape $m \times n$
- softmax

output vector of length n

target vector of length n

learning weights such that both vectors **are identical**

→ **weights*[:,i]*** of the **hidden layer** make up the embedding vector of ***token i***

LLM & Transformer:

***C**ontinuous **B**ag **O**f **W**ords*

n:	**number of unique token from corpus**
m:	**desired number of dimensions for embedding**

The competent services have not included them in the agenda on the grounds that they…

target: *included*

context window: *services have not them in the*

one – hot encoding

shallow ANN:

- input layer of shape $n \times m$
- hidden layer of shape $m \times n$
- softmax

output vector of length n

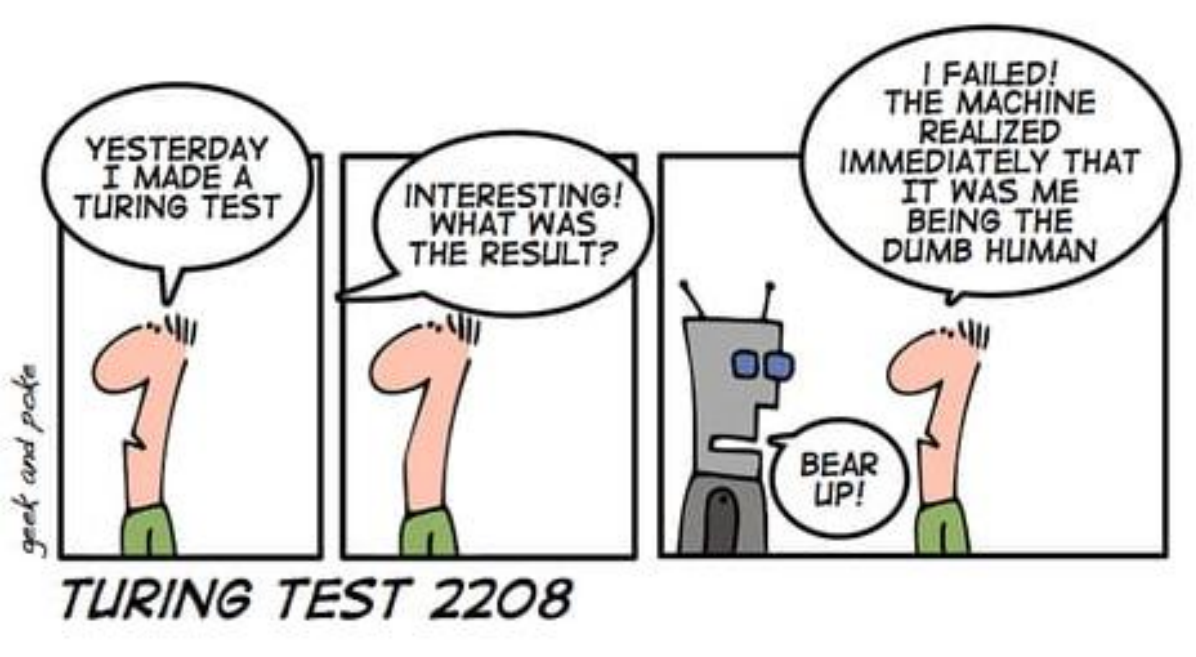
… and so on….

one – hot encoding

target vector of length n

learning weights such that both vectors **are identical**

→ **weights*[:,i]*** of the **hidden layer** make up the embedding vector of ***token i***



Outline

- Introduction
- Bigram and MAP
- Positional Encoding
- Word Embedding
- **Attention**
- Transformer Architecture

LLM & Transformer: Attention

three things make context:

- **positional encoding** (location of token in a sequence)
- **word embedding** (relation between similar/different token)
- **attention** (relation between token within a sequence)

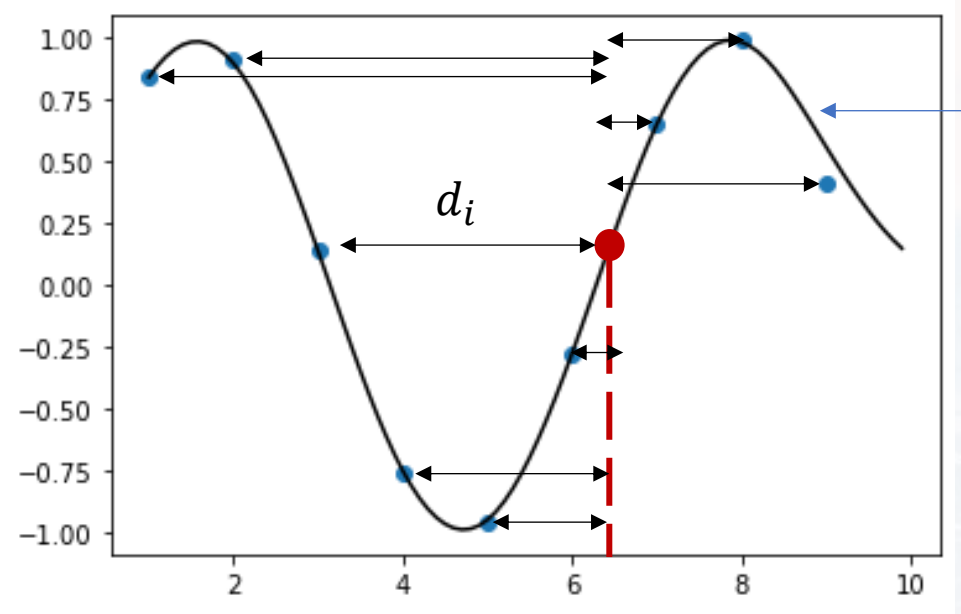
"The cat jumped on the roof."

how the first token influences all other token

how the second token influences all other token

…. and so on

LLM & Transformer: Attention



We want to interpolate between the blue dots
→ generating the black line
→ **no curve fitting!**

<u>idea:</u>

- select a point for which we want the interpolation for
- calculate distance d_i to every other point
- each data point should influence the value of the interpolated point
- the closer, the stronger the influence → weighted mean

$$y_{int} \sim \sum_{i=1}^{I} w_i \, y_i \qquad w_i \sim \frac{1}{\boxed{d_i}}$$

LLM & Transformer:

calculating distance

```
D = np.tile(V, (1, len(V))) -  np.tile(V.transpose(), (len(V), 1))
```

- each data point should influence the value of the interpolated point
- the closer, the stronger the influence → weighted mean

$$y_{int} \sim \sum_{i=1}^{I} w_i \, y_i$$

Gaussian kernel

```
W       = np.exp(-(D**2)/(sigma))
W       = W/np.sum(W + 1e-16, axis = 0)
yint    = np.dot(W.transpose(), y)
```

LLM & Transformer:


```
D = np.tile(V, (1, len(V))) -  np.tile(L.transpose(), (len(V), 1))
```

Gaussian kernel

```
W      = np.exp(-(D**2)/(sigma))
W      = W/np.sum(W + 1e-16, axis = 0)
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```
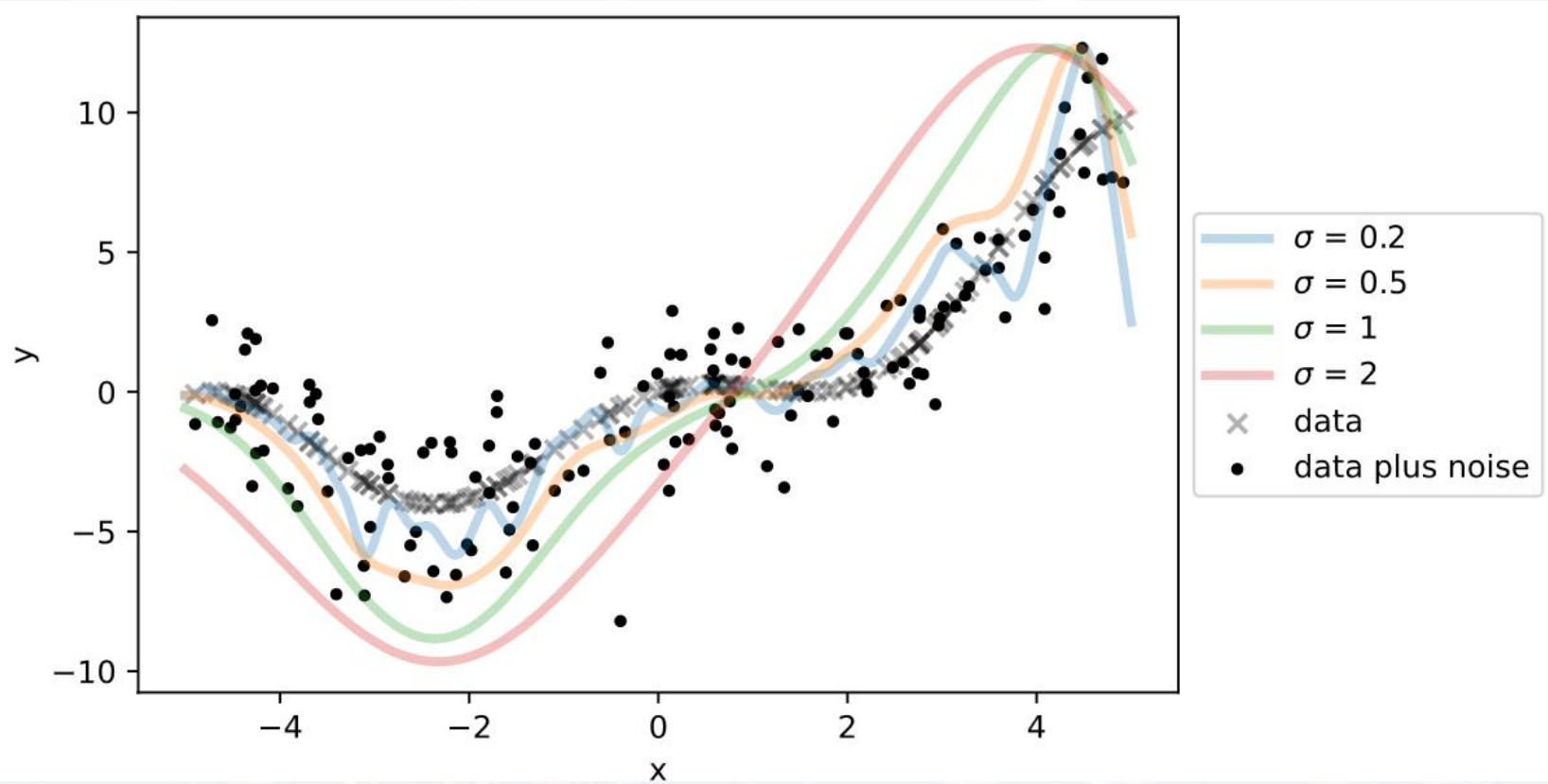
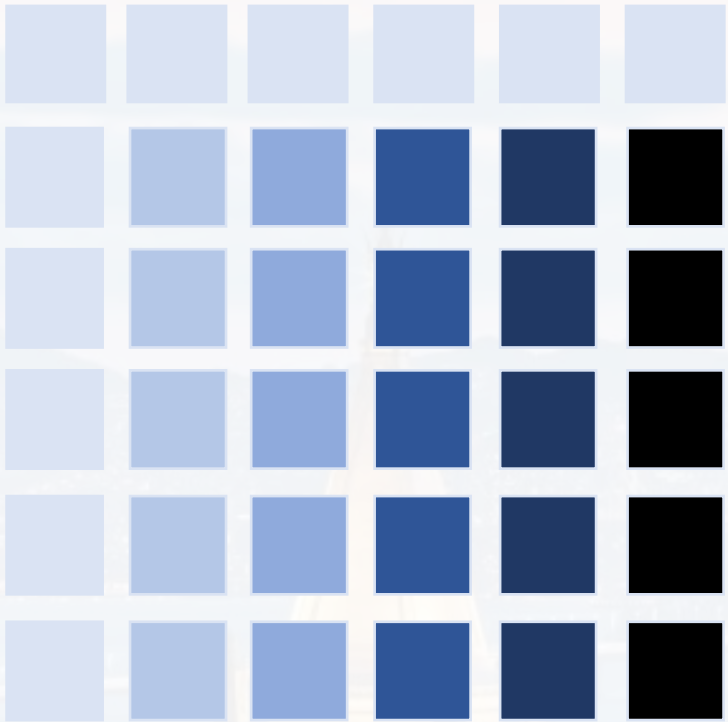



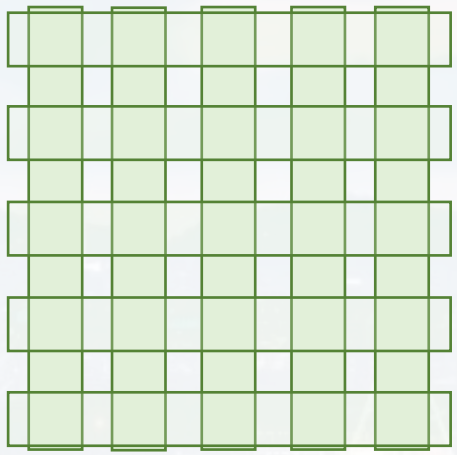
check out:

SmoothGaussKernel.py
SmoothExamples.py

LLM & Transformer:

"The cat jumped on the roof."





Gaussian kernel

```
W     = np.exp(-(D**2)/(sigma))
W     = W/np.sum(W + 1e-16, axis = 0)
yint  = np.dot(W.transpose(), y)
```

actual attention:
these weights are learnable,
no kernel assumed!

LLM & Transformer:

self attention imagine you want to built & train a movie GenAI **that creates movies based on queries**.

training data

"Thrilling horror science fiction movie, plays in space in a distant future"

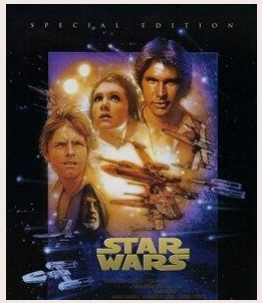
$$X = X_1, X_2, \ldots . X_N$$

each token is a vector X_n of length E

Alien

"Entertaining science fiction movie, plays in space in a distant past"

Star Wars



"Boring love story, plays on a ship in the past"

Titanic

key

value

LLM & Transformer:

self attention

imagine you want to built & train a movie GenAI **that creates movies based on queries**.

training data

each token is a vector X_n of length E $\quad X = X_1, X_2, \ldots . X_N$

key

Alien

value

"Boring love story, plays in space."

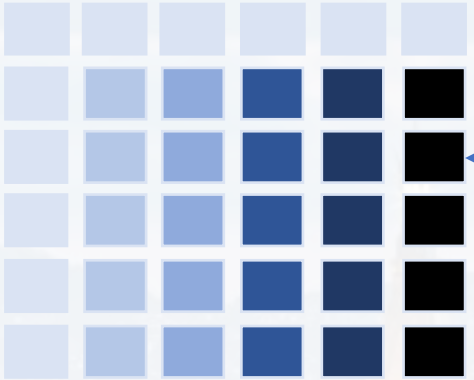
query

LLM & Transformer: Attention

self attention **key** **value** **query**

each token is a vector X_n of length E $X = X_1, X_2, \ldots . X_N$

"Boring love story, plays on a ship in the past"



weights w_{nm}

output:

$$\boxed{Y_n} = \sum_{m=1}^{N} w_{nm} \boxed{X_m}$$

given movie token
(= given y value)

new movie (= new y value)
from weighted mean

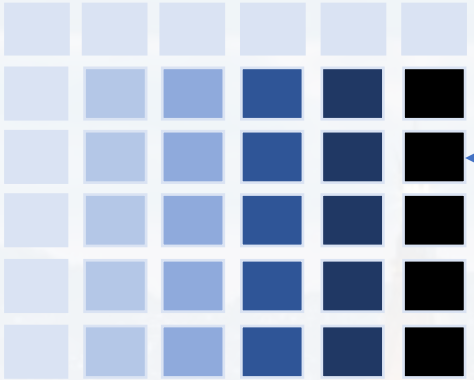
```
W        = np.exp(-(D**2)/(sigma))
W        = W/np.sum(W + 1e-16, axis = 0)
yint     = np.dot(W.transpose(), y)
```

LLM & Transformer: Attention

self attention **key** **value** **query**

each token is a vector X_n of length E $X = X_1, X_2, \ldots . X_N$

"Boring love story, plays on a ship in the past"



weights w_{nm}

output:

$$Y_n = \sum_{m=1}^{N} w_{nm} X_m$$

- weights should be trainable
- weight = 0, X_m has no influence on output
- weights should be positive so that neg weights don't counteract positive weights
- normalization: $\sum_{m=1}^{N} w_{nm} = 1$

for returning the best suggestion:

comparing key vector $\boldsymbol{X_n}$ to query vector $\boldsymbol{X_m}$ via dot product

$$w_{nm} = \frac{\exp(\boldsymbol{X_n} \circ \boldsymbol{X_m})}{\sum_{\mu=1}^{N} \exp(\boldsymbol{X_n} \circ \boldsymbol{X_\mu})}$$

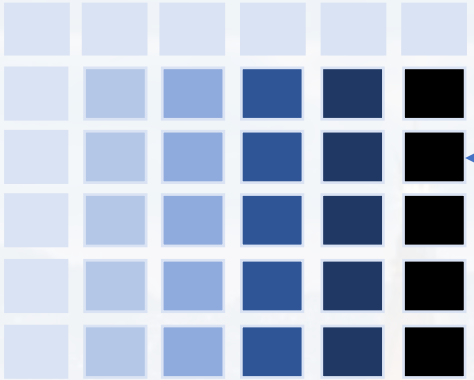
softmax

LLM & Transformer: Attention

self attention **key** **value** **query**

each token is a vector X_n of length E $X = X_1, X_2, \ldots . X_N$

"Boring love story, plays on a ship in the past"



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- normalization: $\sum_{m=1}^{N} w_{nm} = 1$

for returning the best suggestion:

comparing key vector X_n to query vector X_m via dot product

$$w_{nm} = \frac{\exp(X_n \mathrm{W(K)} \circ X_m \mathrm{W(Q)})}{\sum_{\mu=1}^{N} \exp(X_n \mathrm{W(K)} \circ X_\mu \mathrm{W(Q)})}$$

softmax

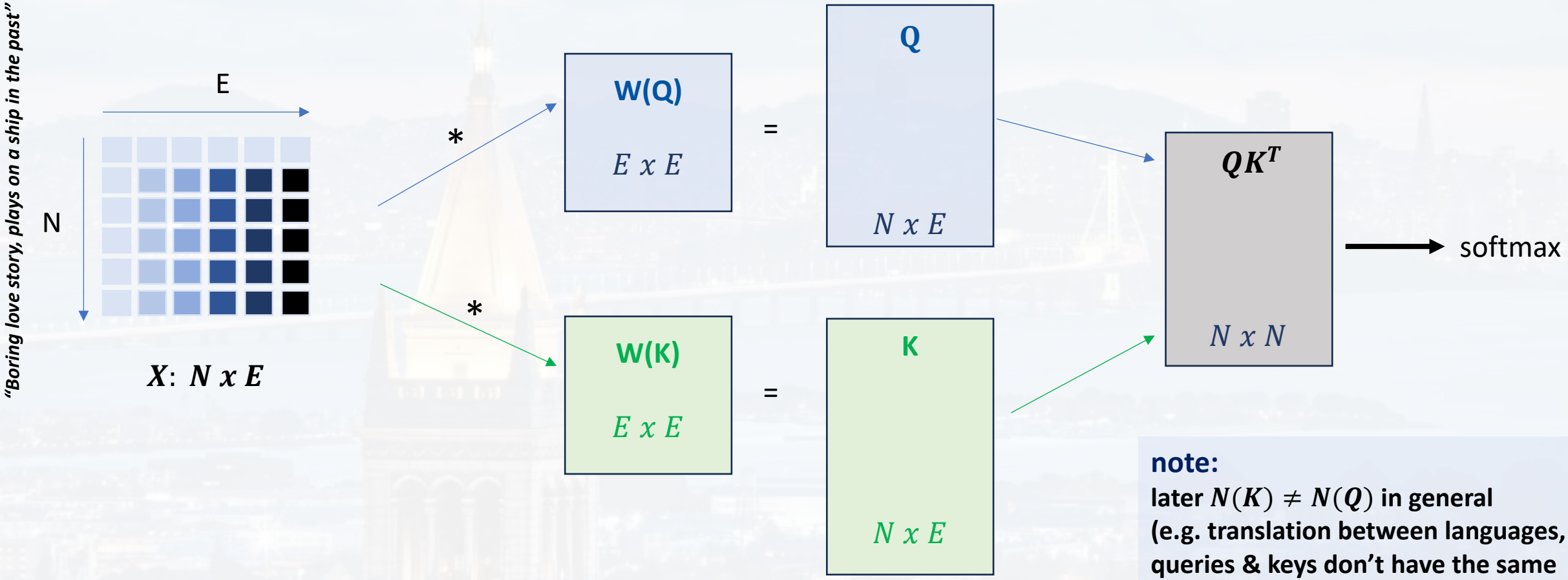
LLM & Transformer: Attention

self attention **key** **value** **query**

$$w_{nm} = \frac{\exp(X_n \mathrm{W(K)} \circ X_m \mathrm{W(Q)})}{\sum_{\mu=1}^{N} \exp(X_n \mathrm{W(K)} \circ X_\mu \mathrm{W(Q)})}$$

N: number of token
E: number of embedding dimensions

output: $$Y_n = \sum_{m=1}^{N} w_{nm} X_m$$



note:
later $N(K) \neq N(Q)$ in general (e.g. translation between languages, queries & keys don't have the same number of tokens in general etc)

LLM & Transformer: Attention

self attention **key** **value** **query**

$$w_{nm} = \frac{\exp(X_n \mathrm{W(K)} \circ X_m \mathrm{W(Q)})}{\sum_{\mu=1}^{N} \exp(X_n \mathrm{W(K)} \circ X_\mu \mathrm{W(Q)})}$$

N: number of token
E: number of embedding dimensions

output: $$Y_n = \sum_{m=1}^{N} w_{nm} X_m = \sum_{m=1}^{N} \frac{\exp(X_n \mathrm{W(K)} \circ X_m \mathrm{W(Q)})}{\sum_{\mu=1}^{N} \exp(X_n \mathrm{W(K)} \circ X_\mu \mathrm{W(Q)})} X_m$$

$$= \sum_{m=1}^{N} softmax(\boldsymbol{Q}_n \boldsymbol{K}_m{}^{\boldsymbol{T}}) X_m$$

given movie (= given y value)

new movie (= new y value)
from weighted mean

$$\rightarrow \sum_{m=1}^{N} softmax(\boldsymbol{Q}_n \boldsymbol{K}_m{}^{\boldsymbol{T}}) X_m \mathrm{W(V)} \qquad \rightarrow \mathrm{Y} = softmax(\boldsymbol{Q}\boldsymbol{K}^{\boldsymbol{T}})\, \mathrm{V}$$ **value**

```
W        = np.exp(-(D**2)/(sigma))
W        = W/np.sum(W + 1e-16, axis = 0)
yint     = np.dot(W.transpose(), y)
```

summarizing the characteristics of the movie by using the movie itself

The output would be a movie, generated by weighted contributions of those movies (values), where the keys match well with the query

LLM & Transformer: Attention

self attention **key** **value** **query**

$$w_{nm} = \frac{\exp(X_n \mathrm{W(K)} \circ X_m \mathrm{W(Q)})}{\sum_{\mu=1}^{N} \exp(X_n \mathrm{W(K)} \circ X_\mu \mathrm{W(Q)})}$$

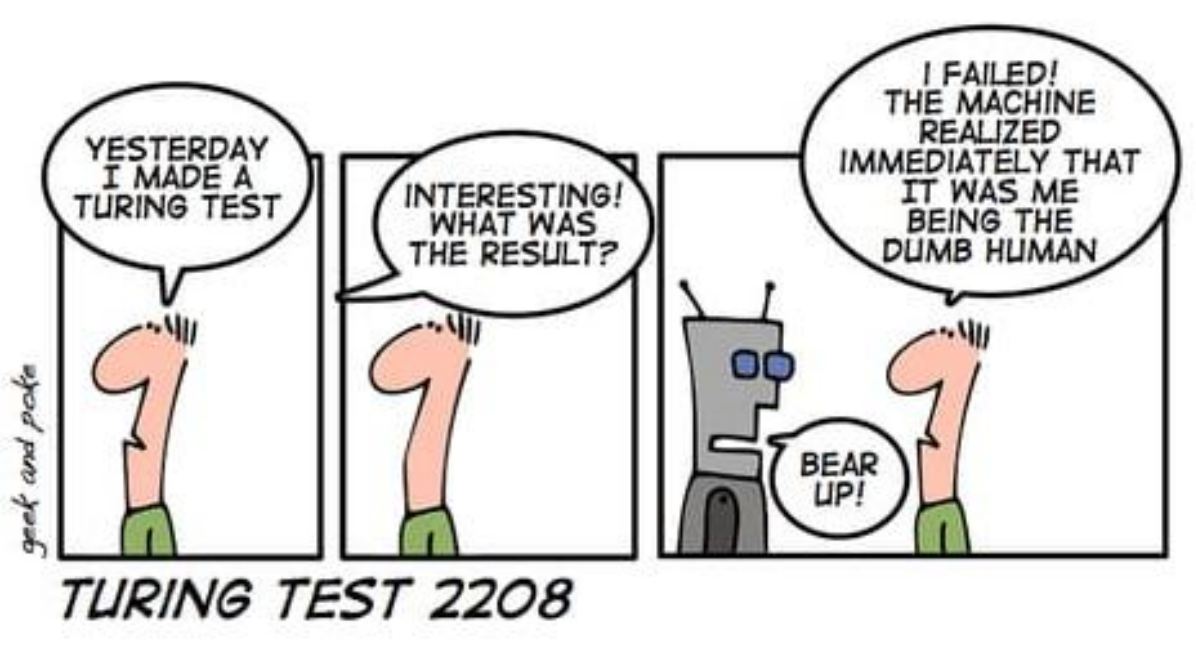
N: number of token
E: number of embedding dimensions

output: $$Y_n = \sum_{m=1}^{N} w_{nm} X_m$$

$$\rightarrow \mathrm{Y} = softmax\left(\frac{QK^T}{\sqrt{E}}\right) \mathrm{V}$$

note:
- dot product scales with E, therefore normalization: $softmax\left(\frac{QK^T}{\sqrt{E}}\right)$
- cross attention:
  - eg. key a phrase in language A, query in language B
  - **encoder/decoder** structure, see next slides

- we want to recognize many underlying pattern → **multiple attention** layers in parallel → transformer



Outline

- Introduction
- Bigram and MAP
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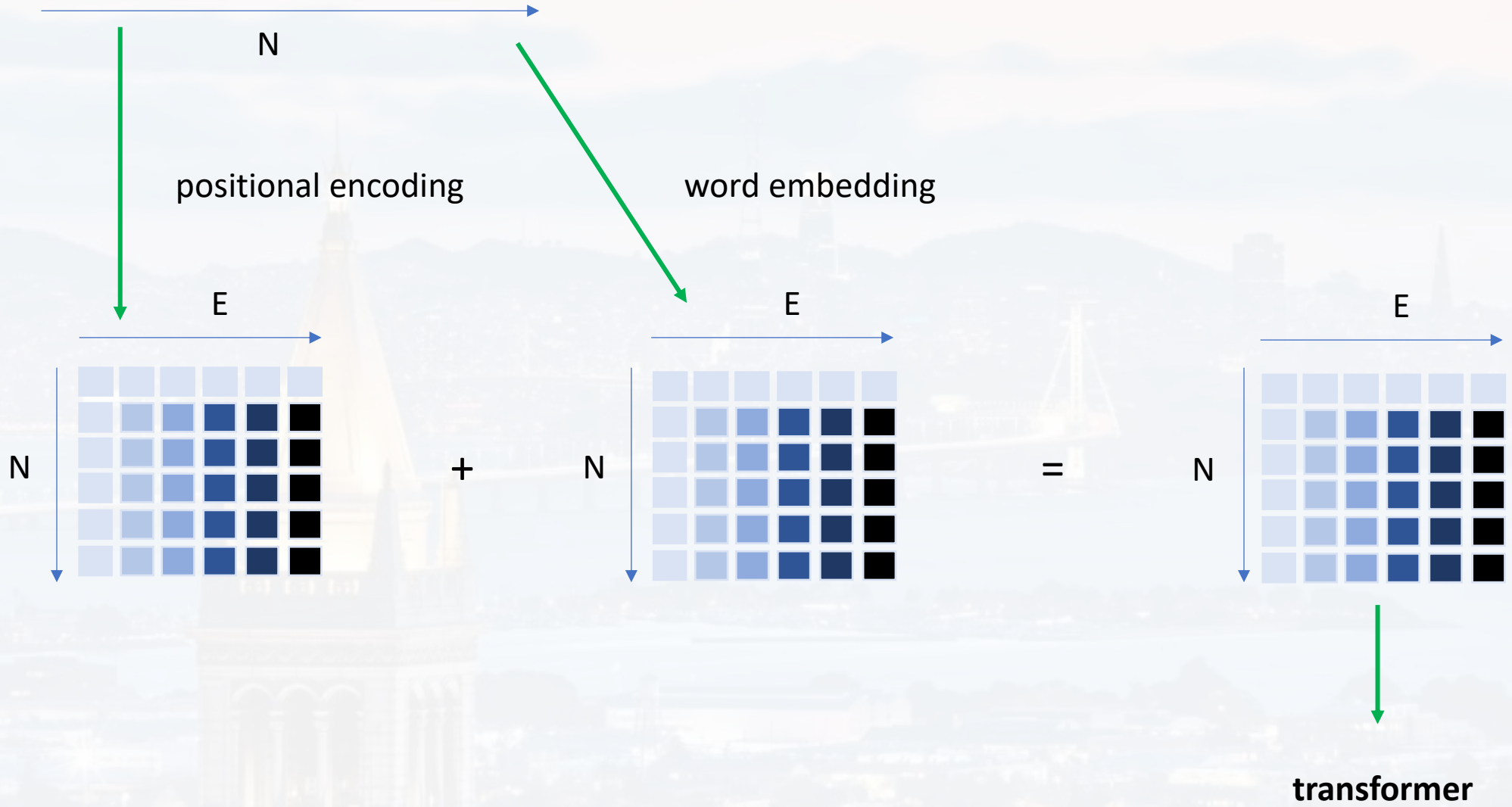
- Word Embedding
- Attention
- **Transformer Architecture**

LLM & Transformer: Transformer Architecture

N: number of token
E: number of embedding dimensions

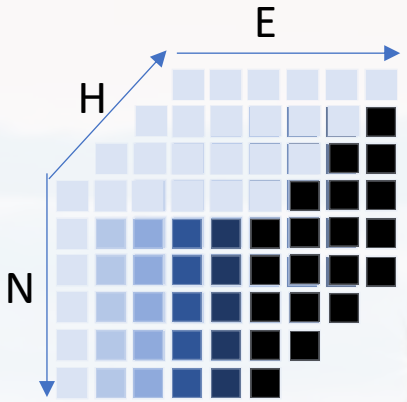
"The cat jumped on the roof."

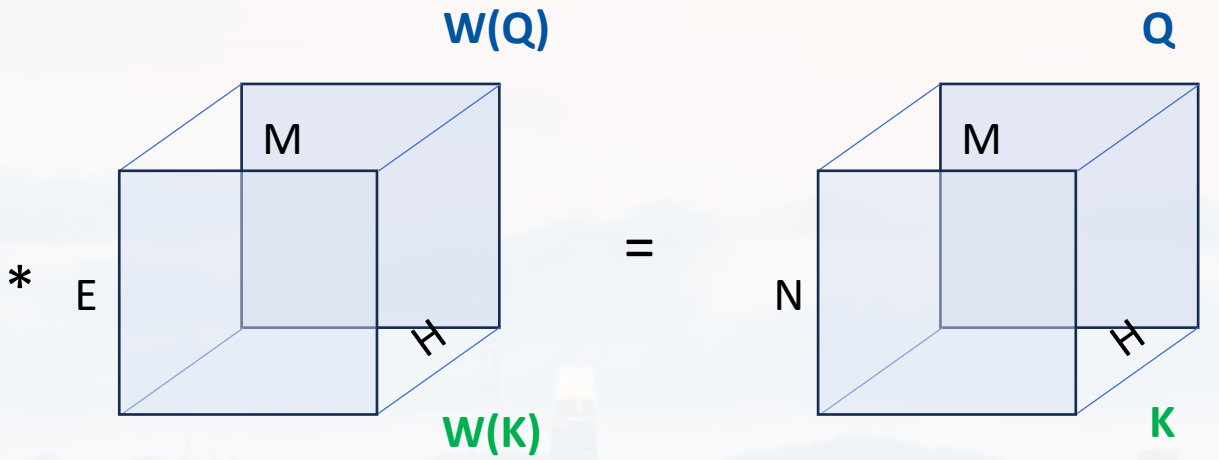


LLM & Transformer:

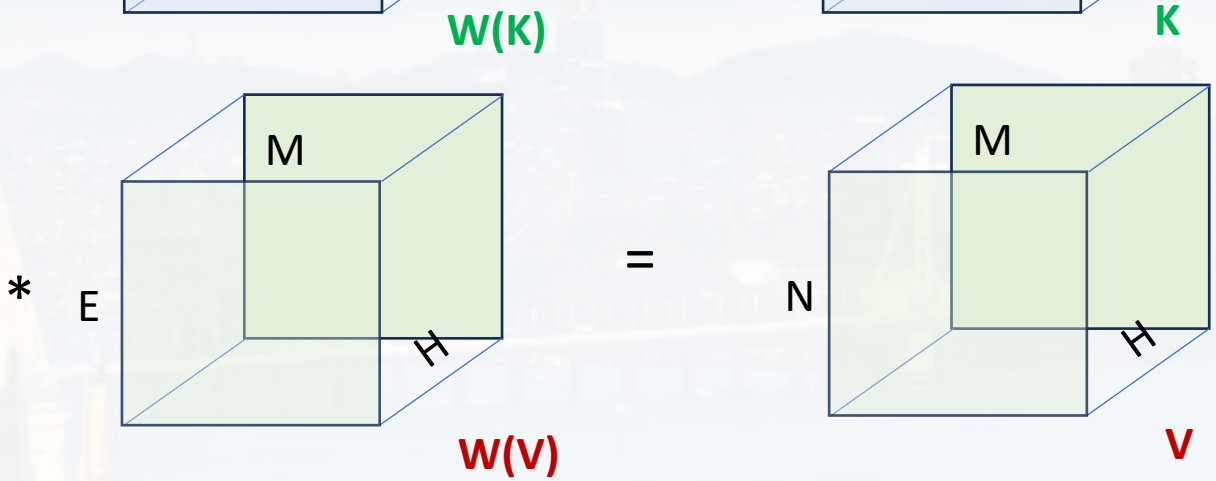
within the transformer:

attention:

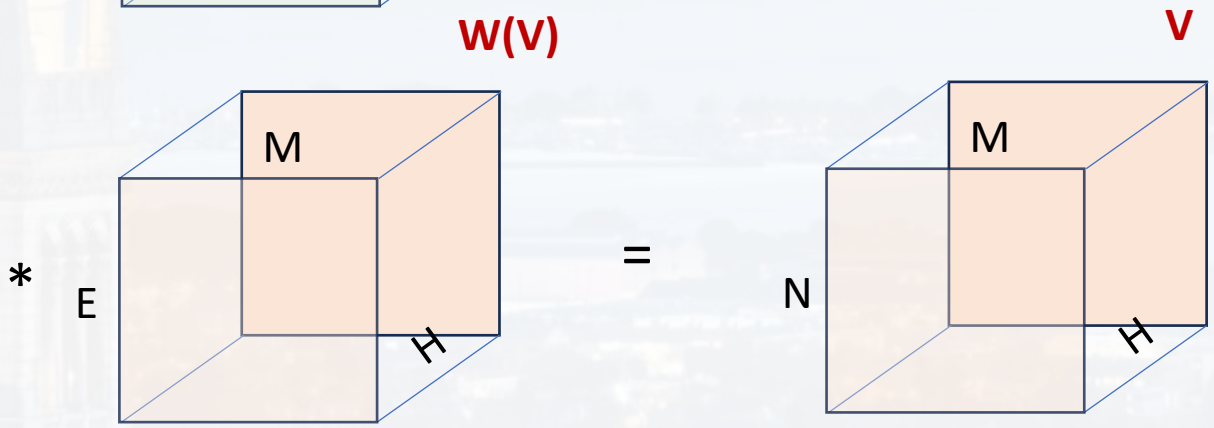




query



key



value

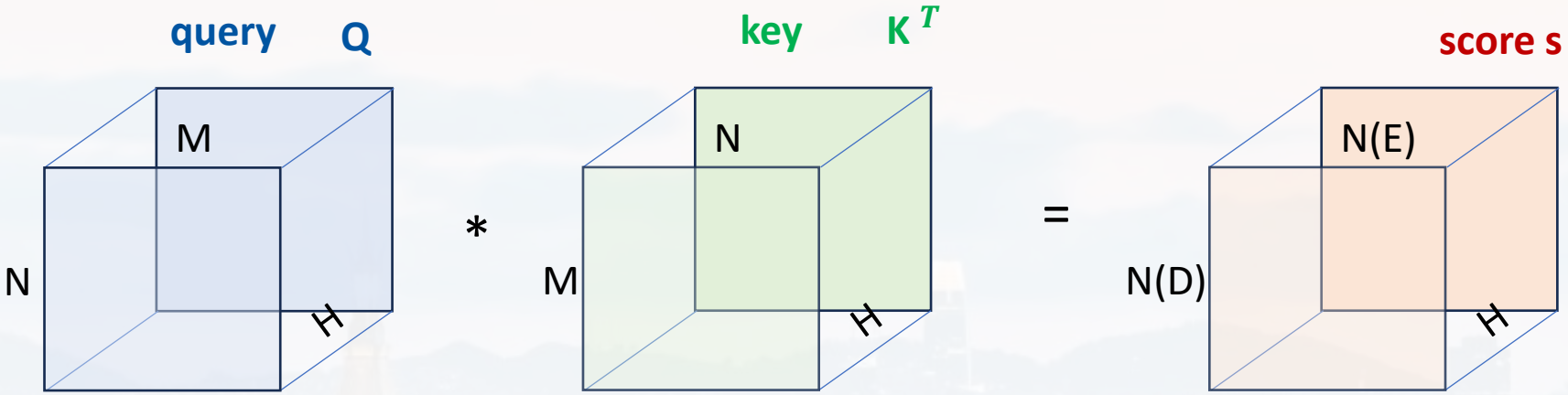
W(Q), W(K), W(V): learnable

N:	**number of token**
E:	**number of embedding dimensions**
H:	**number of heads (= 8)**
M:	**head size (= 64)**

LLM & Transformer:

within the transformer:

attention:

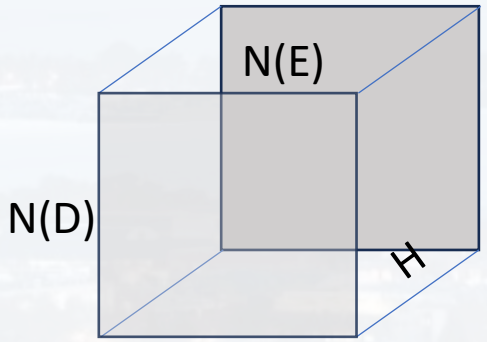


N = N(decoder) N = N(encoder)

if been called by the decoder (see later)

softmax

weights w



W(Q), W(K), W(V): learnable

N:	**number of token**
E:	**number of embedding dimensions**
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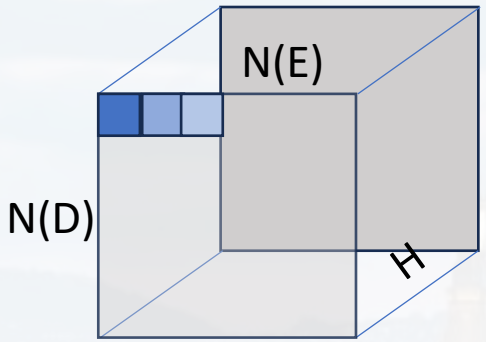
LLM & Transformer:

within the transformer:

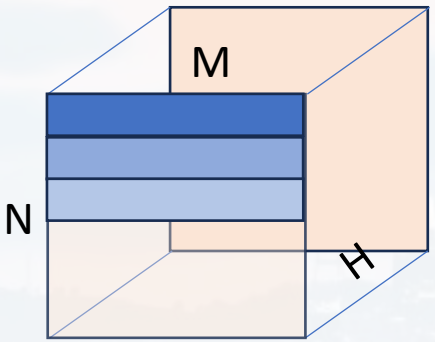
N:	**number of token**
E:	**number of embedding dimensions**
H:	**number of heads (= 8)**
M:	**head size (= 64)**

attention:

weights w



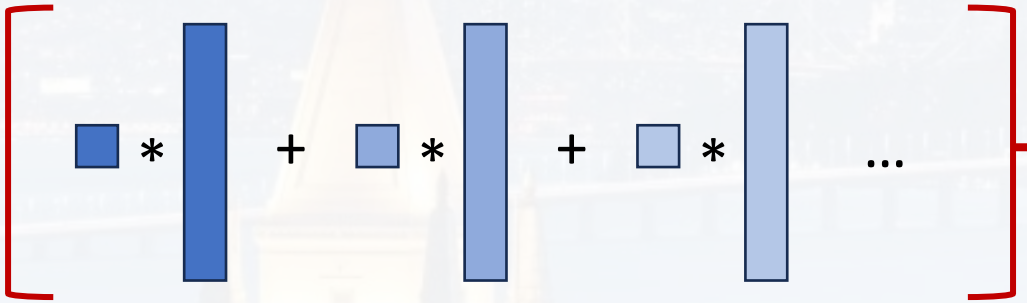
value V



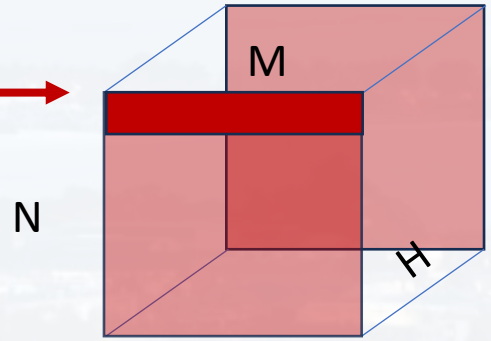
N(E) = N(encoder)

N(D) = N(decoder) , **see later**

$$Y_n = \sum_{m=1}^{N} w_{nm} X_m$$



σ

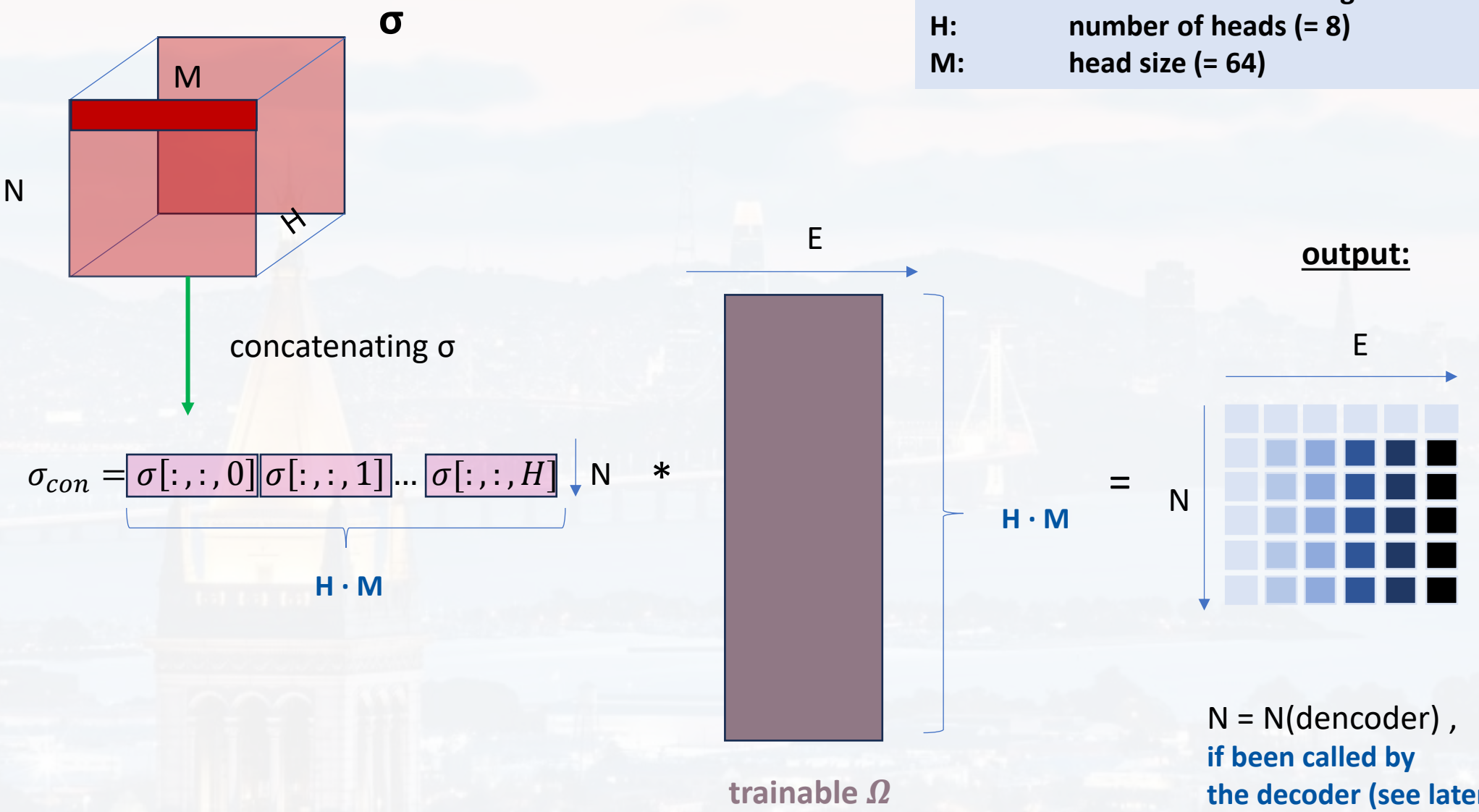


$$\sigma[i,:,k] = \sum_{j} w[i,j,k] * v[j,:,k]$$

LLM & Transformer:

within the transformer:

N:	**number of token**
E:	**number of embedding dimensions**
H:	**number of heads (= 8)**
M:	**head size (= 64)**



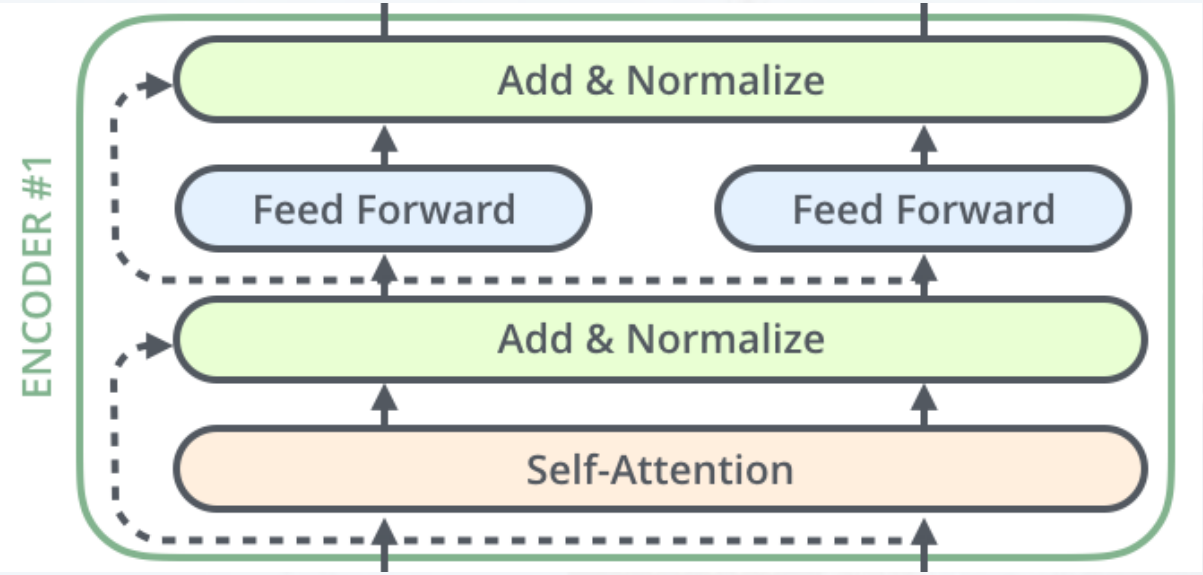
N = N(dencoder) ,
if been called by the decoder (see later)

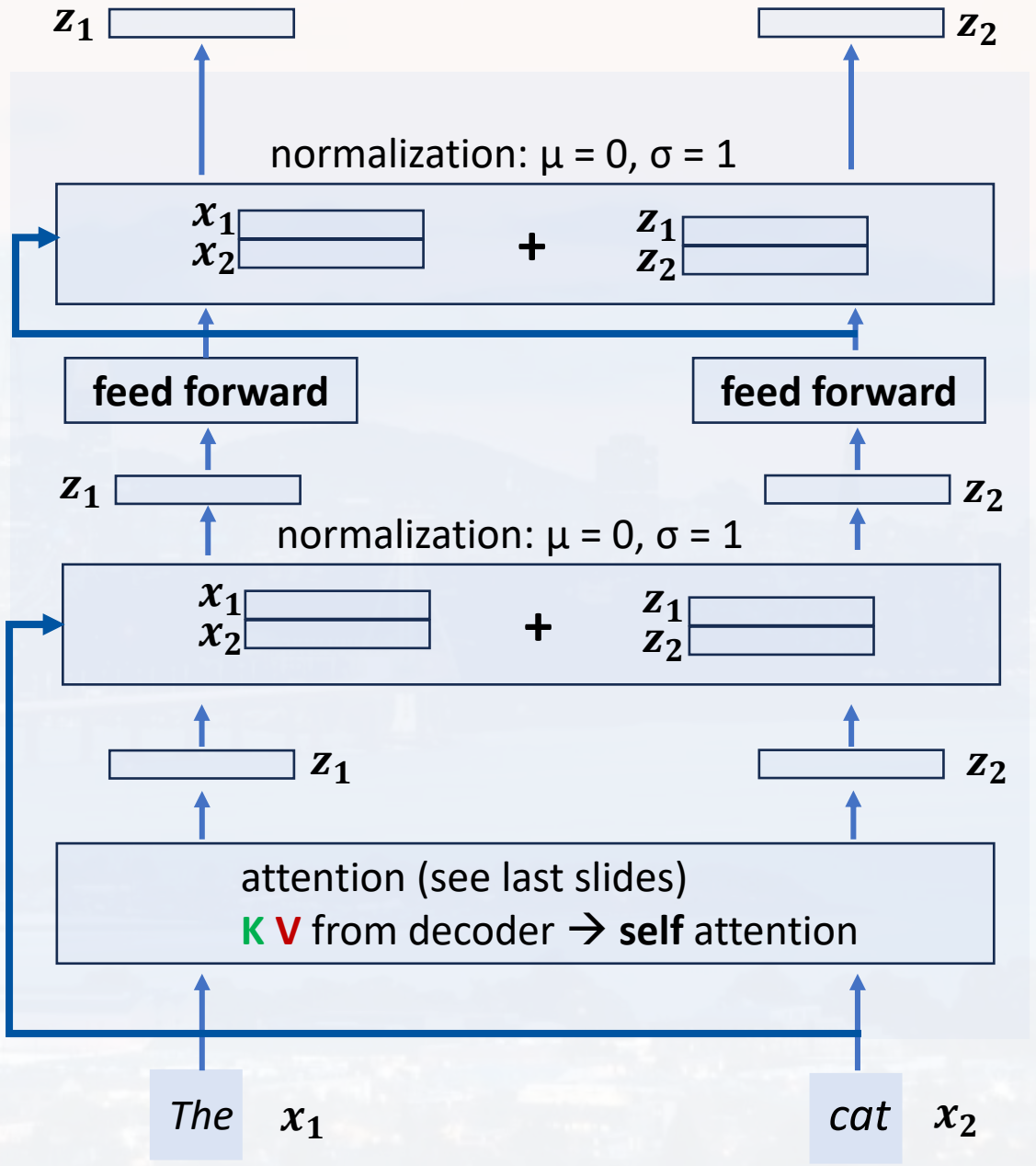
LLM & Transformer:

Transformer Architecture

that was attention → now: encoder

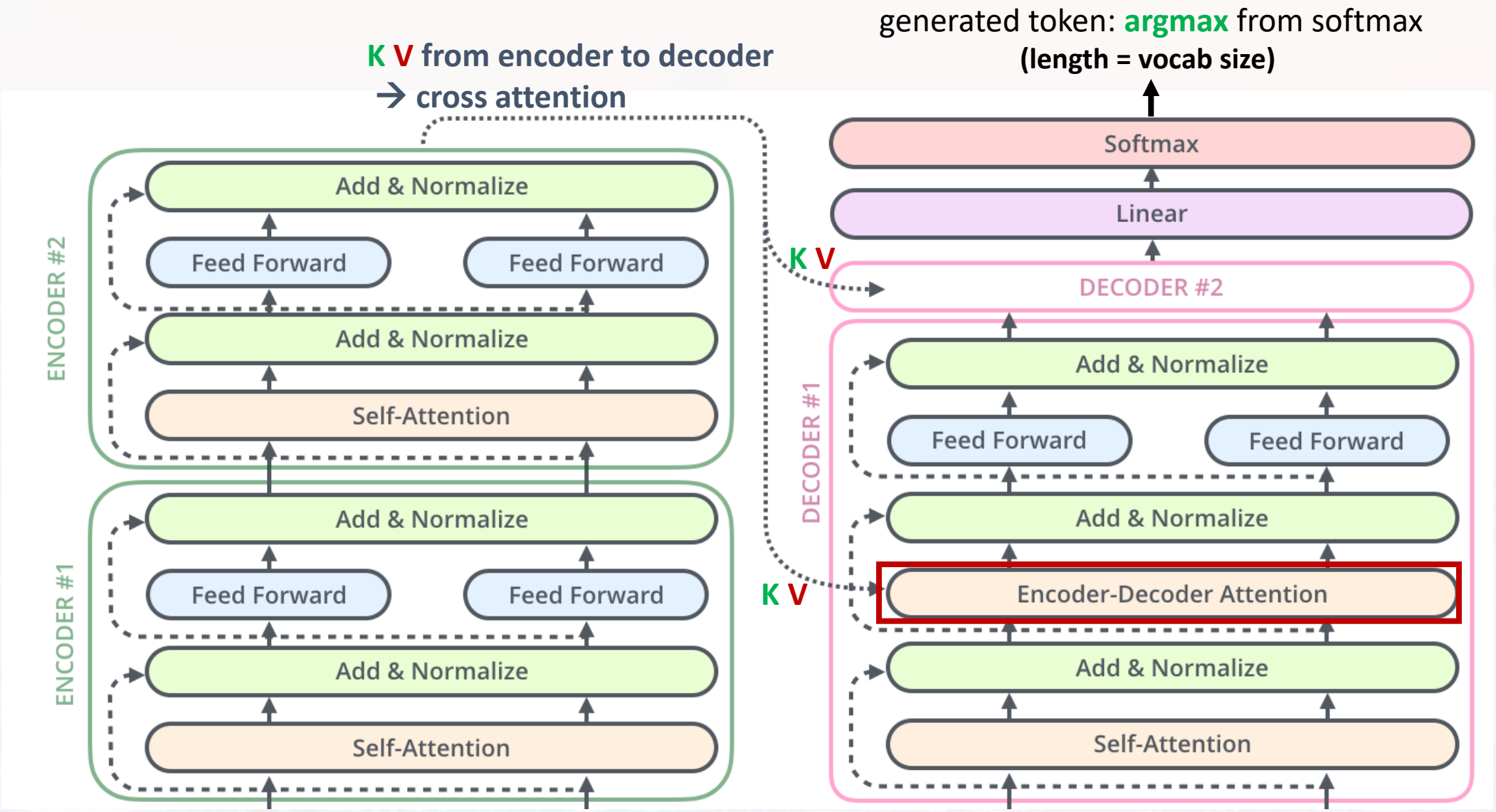
https://jalammar.github.io/illustrated-transformer/





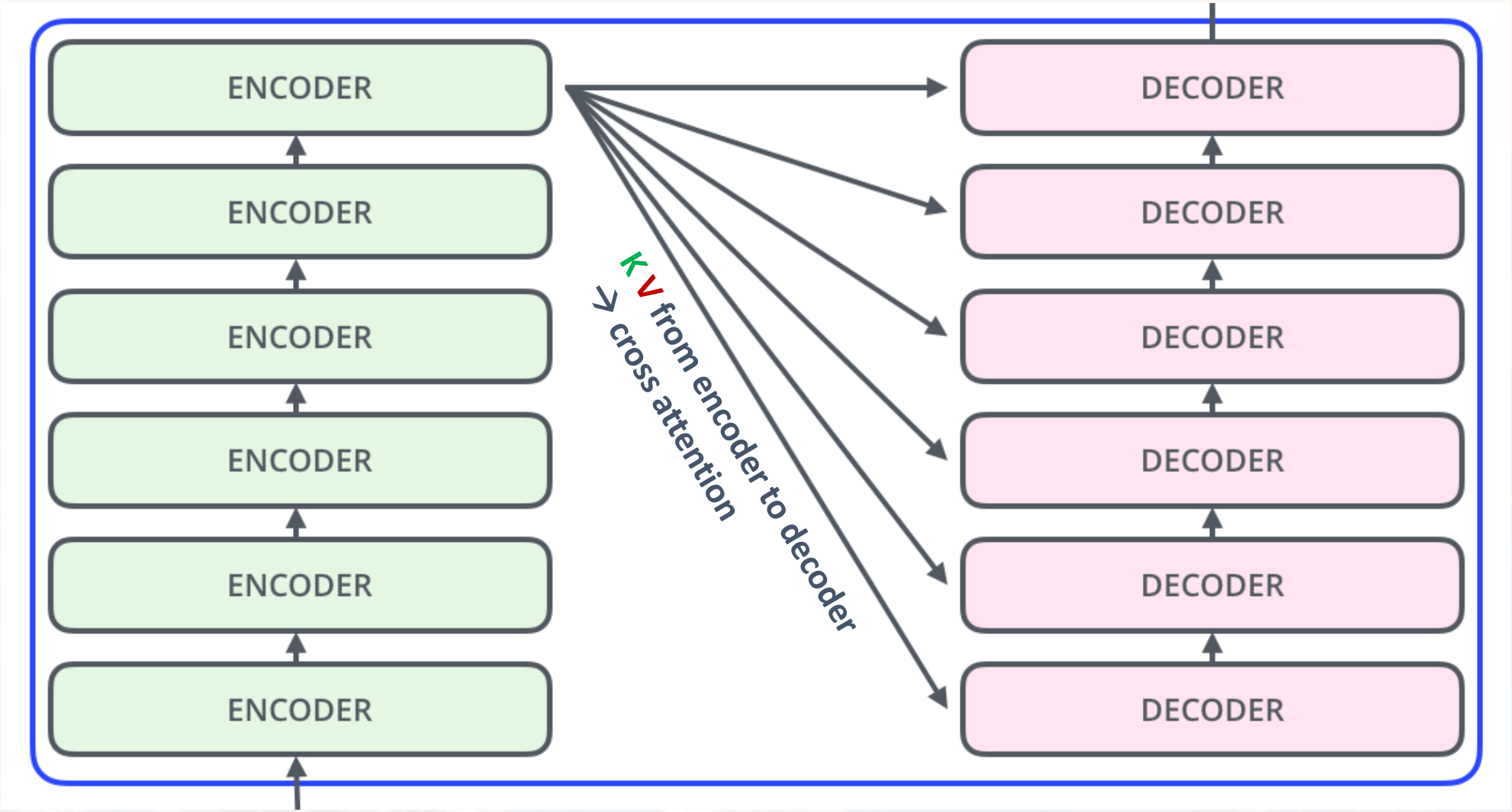
positional encoding + word embedding

LLM & Transformer: Transformer Architecture



https://jalammar.github.io/illustrated-transformer/

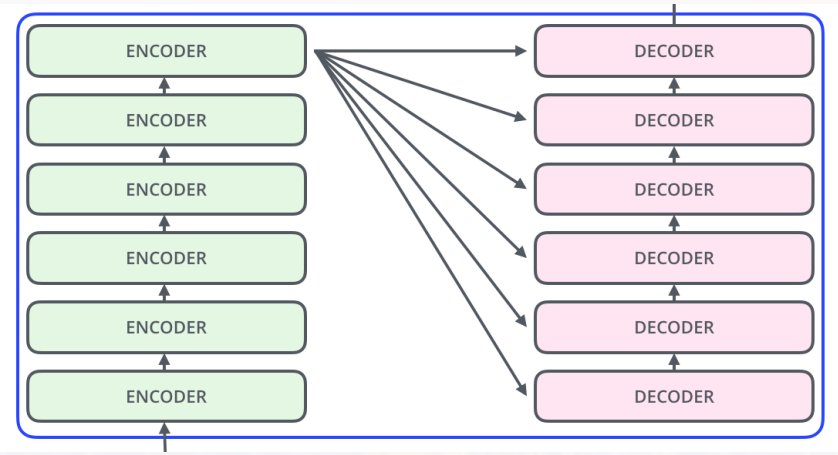
LLM & Transformer: Transformer Architecture



https://jalammar.github.io/illustrated-transformer/

LLM & Transformer:

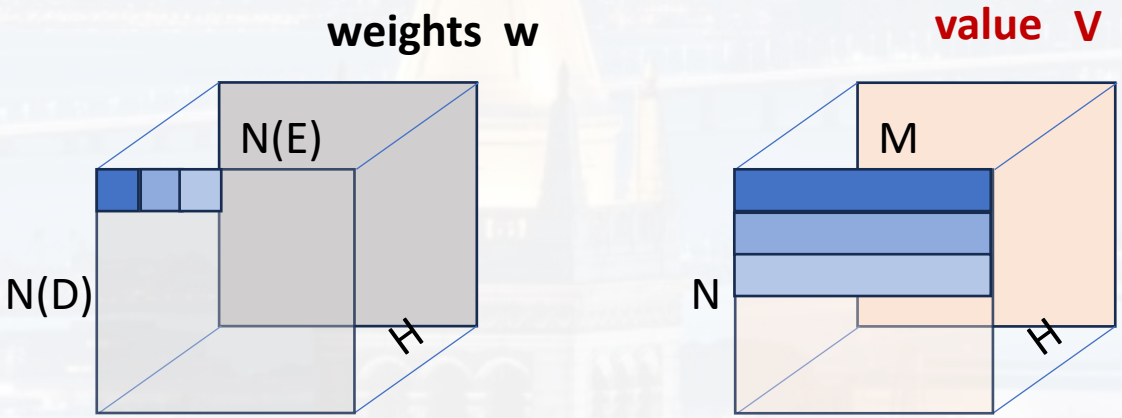
K V from encoder to decoder
→ cross attention



general: *N(E) ≠ N(D)*

English:	Let there be light.	N(E) = 4
German:	Es werde Licht.	N(D) = 3
Latin:	Fiat lux.	N(D) = 2
Hebrew:	yehi ,or	N(D) = 2

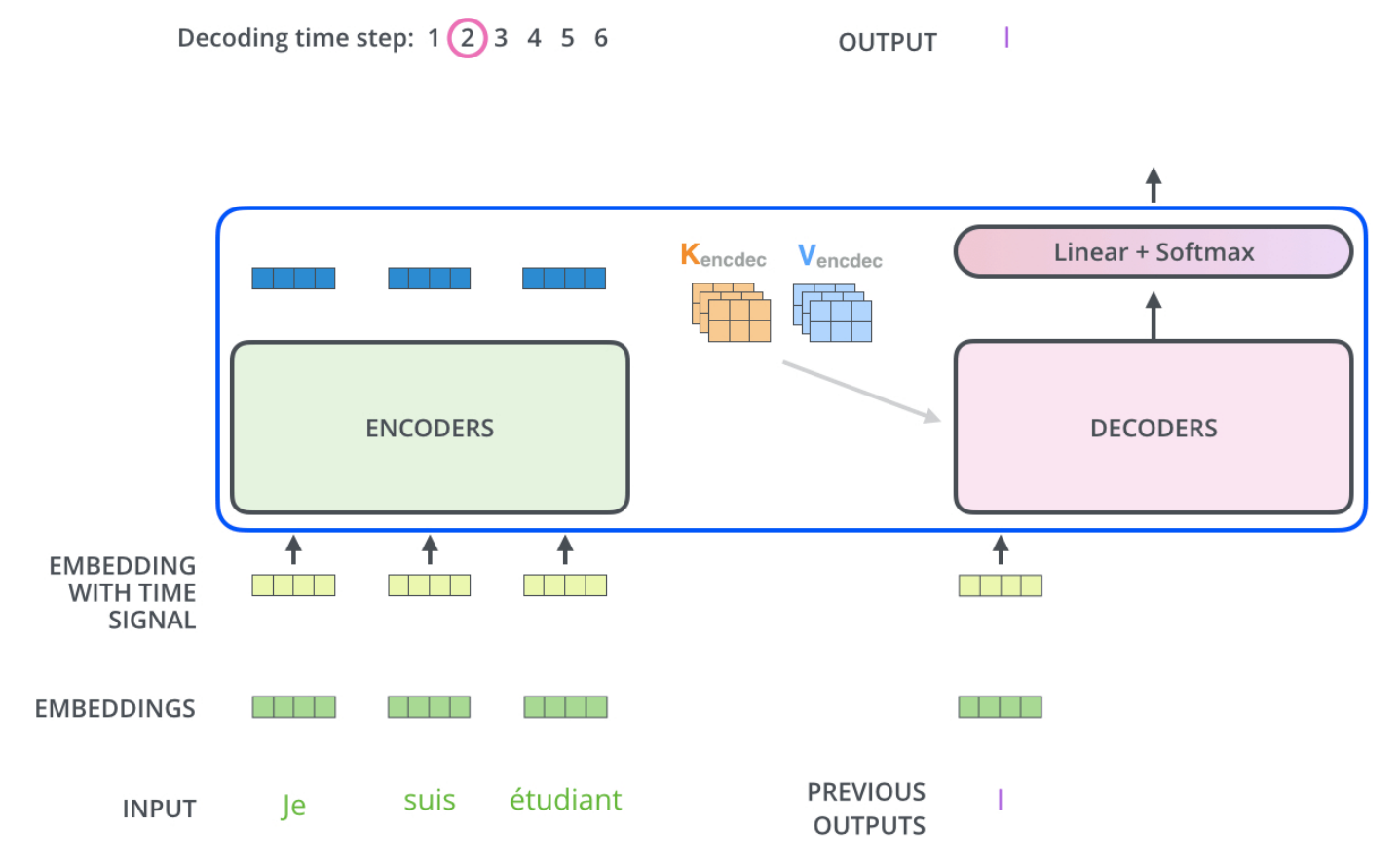
attention:



N = N(encoder) , **if been called by the decoder**

https://jalammar.github.io/illustrated-transformer/

LLM & Transformer: Transformer Architecture



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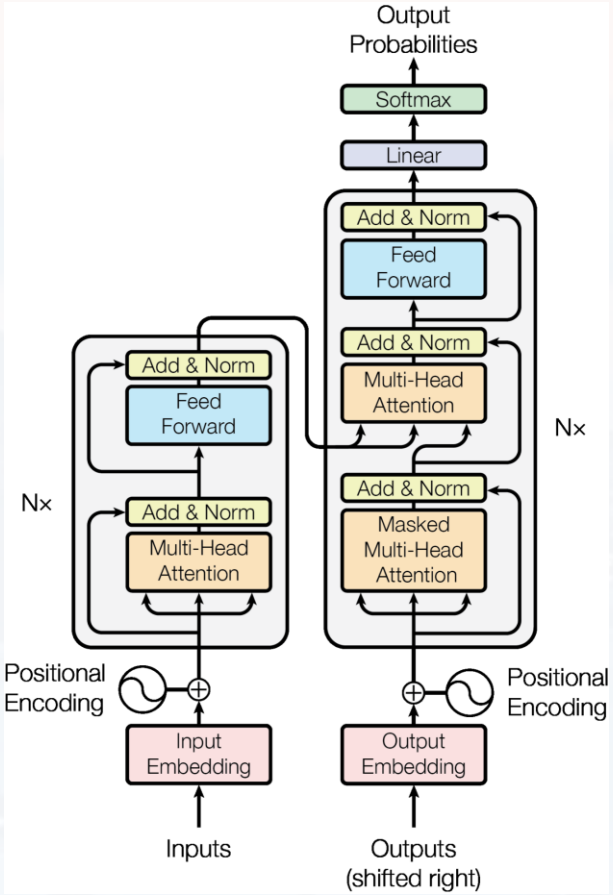


Attention Is All You Need (Vaswani et al, 2017)

LLM & Transformer: Transformer Architecture

mask: $\mathcal{M}$

improvements:

masked attention $\mathrm{Y} = softmax(QK^T)\mathrm{V} \rightarrow \mathrm{Y} = softmax(QK^T + \mathcal{M})\mathrm{V}$

causal masking: $w_{nn+k} = 0 \; \forall \; k > 0$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

token at position t should only consider **previous token** for predicting token at $t + 1$ (natural languages)

padding masking: batches of sequences might have different lengths,
→ shorter sequences are padded with special tokens.
→ model learns to ignore padding tokens
→ for inference

individual masking: we often know that some token can't appear after each other (natural languages)

improvements:

sampling strategies

vanilla: returning most **probable token**, from which we calculate the probabilities for the next token and return the most likely one etc

beam search: we store b (= **beam width**) sequences of length n and then return the **most likely sequence**

$$P(X_1\, X_2\, X_3\, X_4\, X_5 \ldots . X_n) = P(X_n | X_{n-1} \ldots X_1) P(X_{n-1} | X_{n-2} \ldots X_1) \ldots P(X_1)$$

top K-sampling: consider $\boldsymbol{K}$ **most probable token**
→ renormalize their probabilities
→ draw randomly from these K token

$$p_k = \frac{\exp(\pi_k / T)}{\sum_{k=1}^{K} \exp(\pi_k / T)}$$

$\boldsymbol{\pi_k}$: **probability**
$\boldsymbol{p_k}$: **renormalized probability**
$\boldsymbol{T}$: **"softening" parameter**

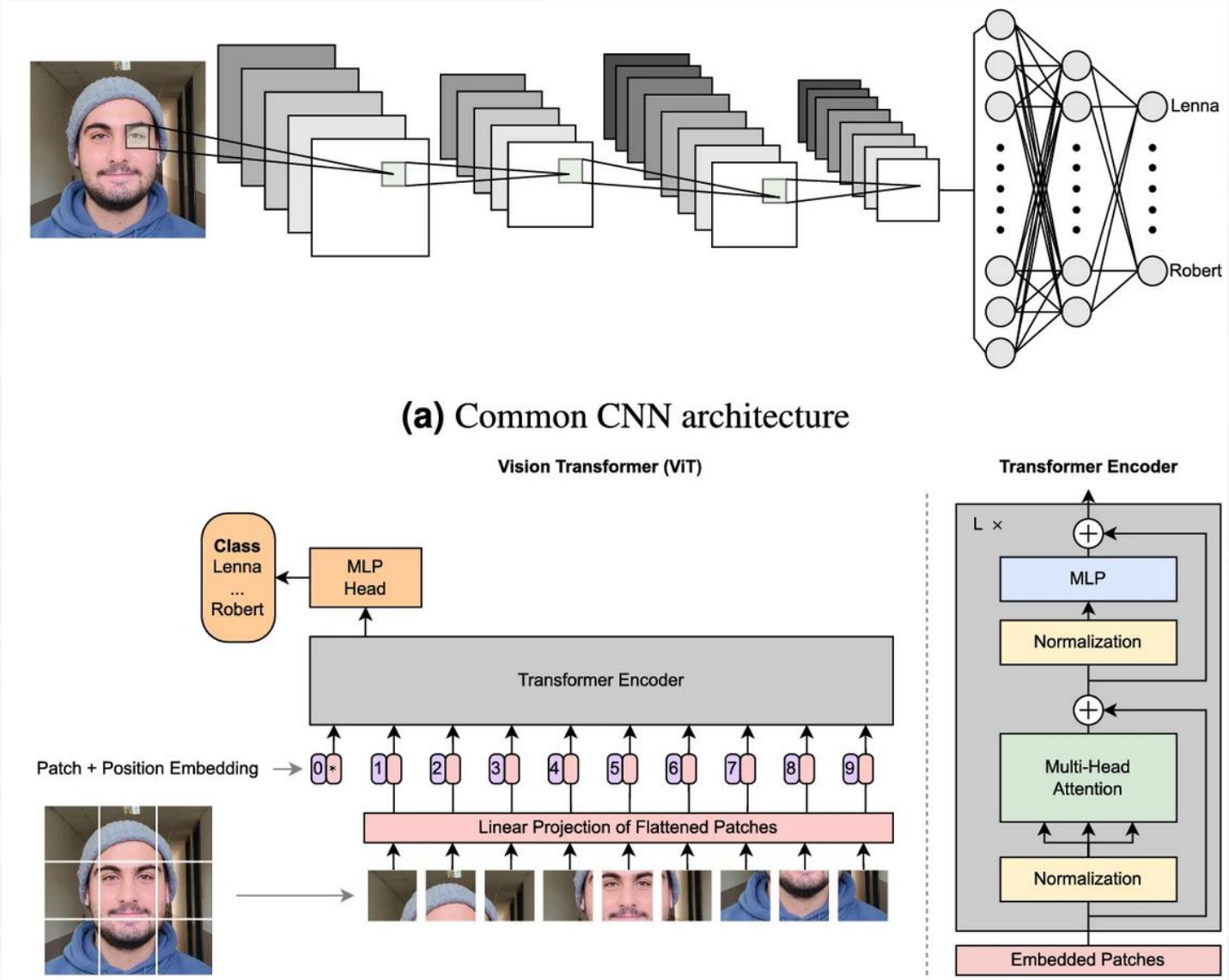
top P-sampling: like top K, but for sequences (see beam search)

LLM & Transformer:

Transformer Architecture

improvements:

vision transformer



Rodrigo, M., Cuevas, C. & García, N. Comprehensive comparison between vision transformers and convolutional neural networks for face recognition tasks. *Sci Rep* 14, 21392 (2024).

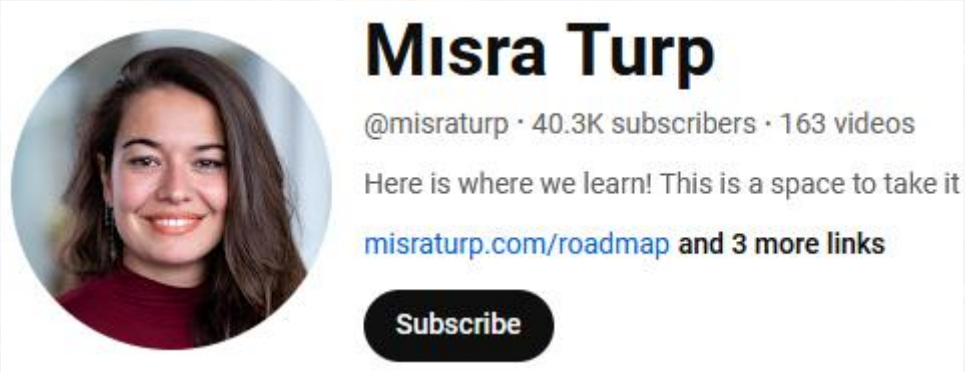
LLM & Transformer:

more about transformers:

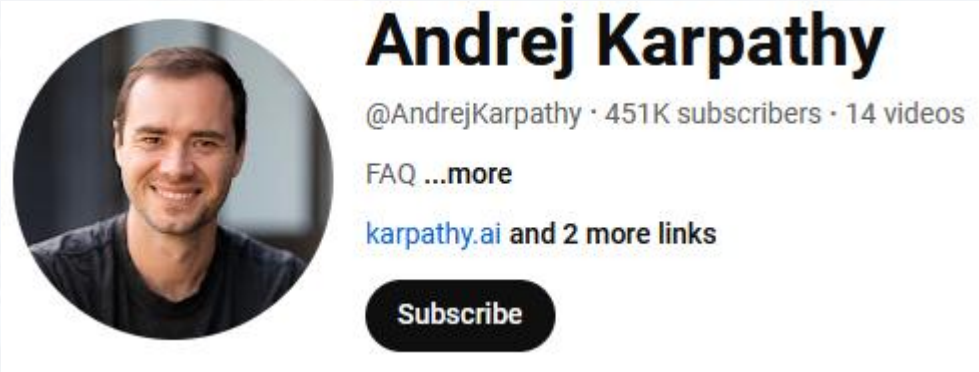
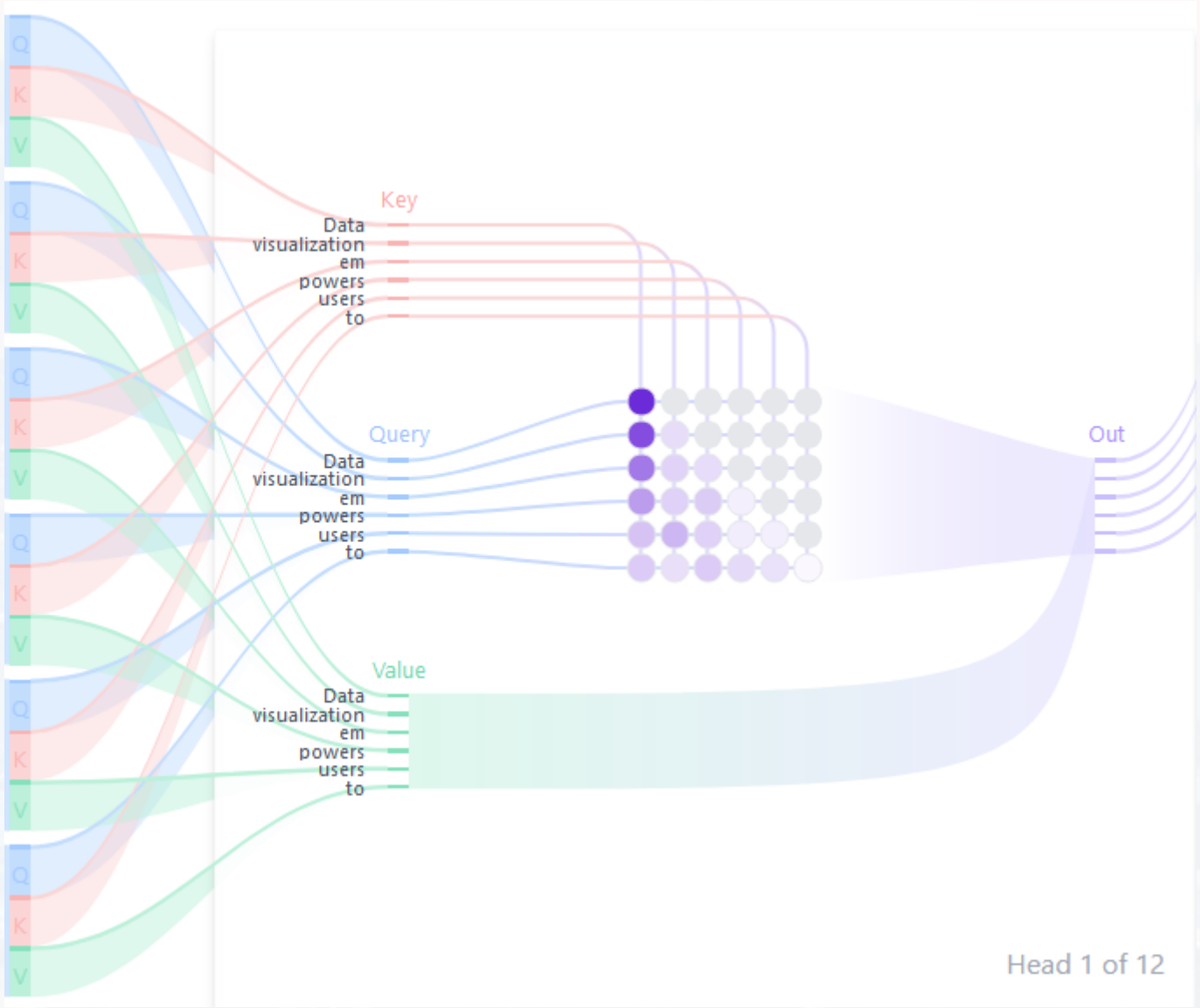
Jay Alammar

Interactive Visualization

transformers intro



building GPT from scratch!



Thank you very much for your